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LLAMA-ADAPTER: EFFICIENT FINE-TUNING OF LARGE LANGUAGE MODELS WITH ZERO-INITIALIZED ATTENTION

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ABSTRACT

With the rising tide of large language models (LLMs), there has been a growing interest in developing general-purpose instruction-following models, e.g., ChatGPT. To this end, we present **LLaMA-Adapter**, a lightweight adaption method for efficient instruction tuning of LLaMA. Using 52K self-instruct demonstrations, LLaMA-Adapter only introduces **1.2M** learnable parameters upon the frozen LLaMA 7B model, and costs less than **one hour** for fine-tuning. Specifically, a zero-initialized attention mechanism is proposed. It adopts a learnable zero gating to adaptively inject the instructional cues into LLaMA within self-attention layers, contributing to a stable training process and superior final performance. In this way, LLaMA-Adapter can generate high-quality responses to diverse language instructions, comparable to Alpaca with fully fine-tuned 7B parameters. Besides language commands, by incorporating an image encoder, our approach can be simply extended to a **Multi-modal LLM** for image-conditioned instruction following, which achieves superior multi-modal reasoning capacity on several popular benchmarks (MME, MMBench, LVLM-eHub). Furthermore, we also verify the proposed zero-initialized attention mechanism for fine-tuning other pre-trained models (ViT, RoBERTa, CLIP) on traditional vision and language tasks, demonstrating the effectiveness and generalizability of our approach. Code and models are released at https://github.com/OpenGVLab/LLaMA-Adapter.

1 INTRODUCTION

Large Language Models (LLMs) (Dai et al., 2019; Radford et al., 2019; Zhang et al., 2022; Raffel et al., 2020; Devlin et al., 2018) have stimulated widespread attention in both academia and industry. Driven by massive corpora and advanced hardware, LLMs exhibit remarkable understanding and generative ability, propelling language tasks to a higher level. Recently, significant progress has been made on instruction-following models, e.g., ChatGPT (OpenAI, 2023a) and GPT-4 (OpenAI, 2023b), which follow language instructions and generate contextual responses. However, the further prevalence of instruction models is largely impeded by the closed-source restriction and high development costs.

To alleviate this, Stanford Alpaca (Taori et al., 2023) proposes to fine-tune an open-source LLM, i.e., LLaMA (Touvron et al., 2023) into an instruction-following model, which is affordable and replicable. Starting from 175 human-written instruction-output pairs (Wang et al., 2022a), Alpaca leverages GPT-3.5 (Brown et al., 2020) to expand the training data to 52K in a self-instruct manner. Supervised by this, Alpaca fine-tunes the entire 7B parameters in LLaMA, producing an exceptional instruction model that performs similarly to GPT-3.5. Despite Alpaca's effectiveness, a complete fine-tuning of large-scale LLaMA is still time-consuming, computation-intensive, and cumbersome to transfer to different downstream scenarios.

* Equal contribution † Corresponding author ‡ Project leader

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LLAMA-ADAPTER：利用零初始化注意力机制实现大型语言模型的高效微调

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摘要

With the rising tide of large language models (LLMs), there has been a growing interest in developing general-purpose instruction-following models, e.g., ChatGPT. To this end, we present **LLaMA-Adapter**, a lightweight adaption method for efficient instruction tuning of LLaMA. Using 52K self-instruct demonstrations, LLaMA-Adapter only introduces **1.2M** learnable parameters upon the frozen LLaMA 7B model, and costs less than **one hour** for fine-tuning. Specifically, a zero-initialized attention mechanism is proposed. It adopts a learnable zero gating to adaptively inject the instructional cues into LLaMA within self-attention layers, contributing to a stable training process and superior final performance. In this way, LLaMA-Adapter can generate high-quality responses to diverse language instructions, comparable to Alpaca with fully fine-tuned 7B parameters. Besides language commands, by incorporating an image encoder, our approach can be simply extended to a **Multi-modal LLM** for image-conditioned instruction following, which achieves superior multi-modal reasoning capacity on several popular benchmarks (MME, MMBench, LVLM-eHub). Furthermore, we also verify the proposed zero-initialized attention mechanism for fine-tuning other pre-trained models (ViT, RoBERTa, CLIP) on traditional vision and language tasks, demonstrating the effectiveness and generalizability of our approach. Code and models are released at https://github.com/OpenGVLab/LLaMA-Adapter.

1 引言

Large Language Models (LLMs) (Dai et al., 2019; Radford et al., 2019; Zhang et al., 2022; Raffel et al., 2020; Devlin et al., 2018) have stimulated widespread attention in both academia and industry. Driven by massive corpora and advanced hardware, LLMs exhibit remarkable understanding and generative ability, propelling language tasks to a higher level. Recently, significant progress has been made on instruction-following models, e.g., ChatGPT (OpenAI, 2023a) and GPT-4 (OpenAI, 2023b), which follow language instructions and generate contextual responses. However, the further prevalence of instruction models is largely impeded by the closed-source restriction and high development costs.

To alleviate this, Stanford Alpaca (Taori et al., 2023) proposes to fine-tune an open-source LLM, i.e., LLaMA (Touvron et al., 2023) into an instruction-following model, which is affordable and replicable. Starting from 175 human-written instruction-output pairs (Wang et al., 2022a), Alpaca leverages GPT-3.5 (Brown et al., 2020) to expand the training data to 52K in a self-instruct manner. Supervised by this, Alpaca fine-tunes the entire 7B parameters in LLaMA, producing an exceptional instruction model that performs similarly to GPT-3.5. Despite Alpaca's effectiveness, a complete fine-tuning of large-scale LLaMA is still time-consuming, computation-intensive, and cumbersome to transfer to different downstream scenarios.

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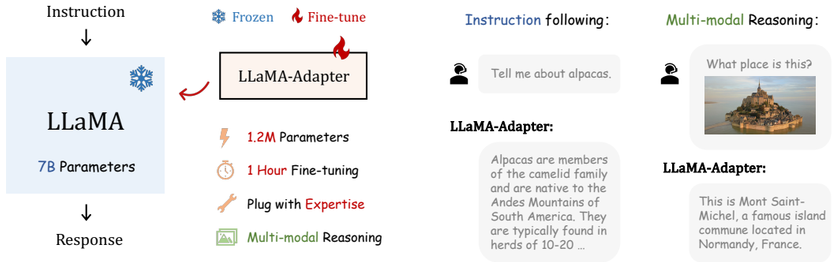


Figure 1: **Characteristics of LLaMA-Adapter.** Our lightweight adaption method efficiently fine-tunes LLaMA (Touvron et al., 2023) 7B model with only 1.2M learnable parameters within one hour, which exhibits superior instruction-following and multi-modal reasoning capacity.

In this paper, we introduce **LLaMA-Adapter**, an efficient fine-tuning method that adapts LLaMA into a well-performed instruction-following model. Trained by Alpaca's instruction-output data, our approach freezes the entire LLaMA model, and proposes a zero-initialized attention mechanism with superior resource efficiency. Specifically, in LLaMA's higher transformer layers, we append a set of learnable adaption prompts as prefixes to the word tokens. Then, to avoid the noise from randomly initialized prompts at the early training stage, we equip the frozen self-attention layers with a learnable gating factor. The gating mechanism is initialized by zeros, and controls the feature interaction between prompt and word tokens, within the process of attention calculation. Such a strategy can first preserve the original knowledge in LLaMA, and progressively inject the new instructional signals during training. This contributes to a more stable learning process and better instruction-following capacity of the final model.

Overall, our LLaMA-Adapter exhibits four main characteristics, as shown in Figure 1.

- **1.2M Parameters.** Instead of updating the full 7B parameters, we freeze the pre-trained LLaMA and only learn the zero-initialized attention mechanism with 1.2M parameters. This, however, reveals comparable instruction-following proficiency with the 7B Alpaca.
- **One-hour Fine-tuning.** Thanks to our lightweight adaption modules with zero-initialized gating, the training convergence of LLaMA-Adapter costs less than one hour on 8 A100 GPUs, which are three times faster than Alpaca.
- **Plug with Expertise.** For different scenarios, it is flexible to insert their respective adapters to endow LLaMA with different expert knowledge or new modality input. Thus, it suffices to store a 1.8M adapter within each context, other than a complete copy of the 13G LLaMA.
- **Multi-modal Reasoning.** Besides language instruction, our approach can also incorporate an image encoder via zero-initialized attention to become a multi-modal LLM. Compared to concurrent works (Liu et al., 2023b; Zhu et al., 2023), LLaMA-Adapter showcases higher tuning efficiency with competitive reasoning capacity on MME (Fu et al., 2023), MMBench (Liu et al., 2023c), and LVLM-eHub (Xu et al., 2023) benchmarks.

In addition to instruction tuning, our zero-initialized attention can be generalized to traditional vision and language tasks for parameter-efficient fine-tuning. We apply our approach to the pre-trained ViT (Dosovitskiy et al., 2020), ReBERTa (Liu et al., 2019), and CLIP (Radford et al., 2021), respectively for fine-tuning vision, language, and vision-language models. On a wide range of downstream tasks, we demonstrate the effectiveness of our proposed method for traditional tasks.

2 Related Work

Instruction Tuning of Language Models. The subfield of language models learning instruction-following capabilities aims to generate responses based on natural language commands. These methods normally enhance the pre-trained LLMs by fine-tuning them with high-quality instruction-output data pairs. Early works, such as FLAN (Wei et al., 2021), PromptSource (Bach et al.,



图1：**LLaMA-Adapter的特点**。我们的轻量级适配方法仅需一小时，即可使用仅含120万个可学习参数的70亿参数LLaMA（Touvron等人，2023年）高效完成微调，从而使该模型具备更出色的指令遵循能力与多模态推理能力。

In this paper, we introduce **LLaMA-Adapter**, an efficient fine-tuning method that adapts LLaMA into a well-performed instruction-following model. Trained by Alpaca's instruction-output data, our approach freezes the entire LLaMA model, and proposes a zero-initialized attention mechanism with superior resource efficiency. Specifically, in LLaMA's higher transformer layers, we append a set of learnable adaption prompts as prefixes to the word tokens. Then, to avoid the noise from randomly initialized prompts at the early training stage, we equip the frozen self-attention layers with a learnable gating factor. The gating mechanism is initialized by zeros, and controls the feature interaction between prompt and word tokens, within the process of attention calculation. Such a strategy can first preserve the original knowledge in LLaMA, and progressively inject the new instructional signals during training. This contributes to a more stable learning process and better instruction-following capacity of the final model.

总体而言，如图1所示，我们的LLaMA-Adapter具有四大主要特征。

- **1120万参数**。我们并未更新全部70亿参数，而是冻结预训练好的LLaMA，仅学习拥有120万参数的零初始化注意力机制。尽管如此，该模型的指令遵循能力依然能与70亿参数的Alpaca模型相媲美。
- **一小时微调**。得益于我们那些采用零初始化门控机制的轻量级适配模块，LLLaMA-Adapter在8块A100 GPU上的训练收敛时间仅需不到一小时，这一速度是Alpaca模型的三倍。
- **专家知识整合**。针对不同的应用场景，可以灵活地插入相应的适配器，从而为LLaMA赋予不同的专家知识或新的模态输入。因此，除了存储完整的130亿参数版LLaMA外，只需在每个上下文环境中保存一个180万的适配器即可。
- **Multi-modal Reasoning.** Besides language instruction, our approach can also incorporate an image encoder via zero-initialized attention to become a multi-modal LLM. Compared to concurrent works (Liu et al., 2023b; Zhu et al., 2023), LLaMA-Adapter showcases higher tuning efficiency with competitive reasoning capacity on MME (Fu et al., 2023), MMBench (Liu et al., 2023c), and LVLM-eHub (Xu et al., 2023) benchmarks.

除了进行指令微调之外，我们所采用的零初始化注意力机制还可被推广应用于传统的视觉与语言任务，从而实现参数效率更高的微调。我们将该方法分别应用于预训练的ViT模型（Dosovitskiy等人，2020年）、ReBERTa模型（Liu等人，2019年）以及CLIP模型（Radford等人，2021年），以此对视觉模型、语言模型以及视觉语言模型进行微调。在多种下游任务中，我们的研究证明了该方法在处理传统任务时的有效性。

2 相关工作

语言模型的指令微调。语言模型中用于学习指令遵循能力的子领域，旨在根据自然语言指令生成响应。这些方法通常通过使用高质量的指令-响应数据对对预训练好的大语言模型进行微调，从而提升其性能。早期的相关研究包括FLAN（Wei等人。、2021年）、PromptSource（Bach等人。）等。

2022), and SUP-NATINST (Wang et al., 2022b), introduce effective instruction tuning methods and establish comprehensive evaluation benchmarks. InstructGPT (Ouyang et al., 2022) demonstrates significant improvement in the instruction-following power, but is closed-source to the community. To promote the open source of instruction models, Stanford Alpaca (Taori et al., 2023) fine-tunes all the 7B parameters of LLaMA (Touvron et al., 2023) with 52K self-instruct data. However, this full-model fine-tuning can be inefficient in both time and computation resources, limiting its transferability to downstream applications. In this paper, we propose LLaMA-Adapter to fine-tune only lightweight zero-initialized attention mechanisms on top of the frozen LLaMA, other than updating parameters of the entire model. There are several works ***concurrent*** to ours, Alpaca-LoRA (alp, 2023), Vicuna (Chiang et al., 2023), and LLaMA-GPT4 (Peng et al., 2023), which aim to improve Alpaca from different aspects. Alpaca-LoRA utilizes the existing LoRA (Hu et al., 2021) to efficiently fine-tune LLaMA, which is restricted to the original network structure and cannot be extended for image input. In contrast, our LLaMA-Adapter achieves higher training efficiency and can be simply generalized to a multi-modal LLM via zero-initialized attention. Vicuna and LLaMA-GPT4 target at constructing a more advanced instruction dataset using ChatGPT (OpenAI, 2023a) and GPT-4 (OpenAI, 2023b), instead of Alpaca's 52K data, which still adopt full fine-tuning without the potential for multi-modal instruction tuning.

Parameter-efficient Fine-tuning. The pre-training and fine-tuning paradigms have been proven to be highly effective in different language and vision tasks. Compared to full fine-tuning, Parameter-Efficient Fine-Tuning (PEFT) (Paul, 2022) methods freeze most parameters of pre-trained models, and aim to exhibit comparable capabilities on downstream tasks (Wang et al., 2018; Puzikov & Gurevych, 2018). Therein, prompt tuning appends a collection of trainable tokens to pre-trained large models, which are inserted either to the input embeddings (Lester et al., 2021; Liu et al., 2021b) or every intermediate layer (Li & Liang, 2021; Liu et al., 2021a). LoRA (Hu et al., 2021; Zhang et al., 2023d; Hedegaard et al., 2022) introduces trainable rank decomposition matrices into each network weights (Karimi Mahabadi et al., 2021), indicating promising fine-tuning ability on large generative models (Cuenca & Paul, 2023; alp, 2023). Adapters (Houlsby et al., 2019) insert lightweight adaption modules into each block of the transformer and have been extended across numerous domains (Gesmundo & Dean, 2022; Gao et al., 2021; Zhang et al., 2021). Different from previous efforts, we propose the LLaMA-Adapter with zero-initialized attention specially designed for instruction tuning and multi-modal reasoning of LLaMA (Touvron et al., 2023). Some existing works also adopt gating techniques in prompt tuning (Yoo et al., 2023; Goswami et al., 2023), but conduct a naive gated combination of different prompt tokens with randomly initialized factors. Instead, our gating factor learns from zero during training, and is delicately integrated into self-attention layers. Another branch of work applies zero initialization to convolutional networks (Zhao et al., 2021), text-to-image diffusion models (ControlNet (Zhang et al., 2023c)), or vision-language learning (Flamingo (Alayrac et al., 2022)). They are not PEFT methods requiring large-scale parameters, and have motivations for better network-level initialization or feature-level fusion via residual connections, very different from our interaction controlling within attention layers.

3 LLaMA-Adapter

In Section 3.1, we first introduce to insert learnable adaption prompts into LLaMA's (Touvron et al., 2023) transformer. Then, we present the details of zero-initialized attention mechanisms with zero gating in Section 3.2, and generalize LLaMA-Adapter for multi-modal reasoning in Section 3.3. Finally, we extend our approach for efficient fine-tuning of language and vision models in Section **??**.

3.1 Learnable Adaption Prompts

Given a pre-trained LLaMA with an N-layer transformer, we first insert a set of learnable adaption prompts into its topmost L layers ($L \leq N$). We denote the prompts as $\{P_l\}_{l=1}^{L}$, where $P_l \in \mathbb{R}^{K \times C}$ with K denoting the prompt length for each layer, and C equaling the feature dimension of LLaMA's transformer. The prompting at last L layers can better tune the language representations with higher-level semantics.

Taking the l-th inserted layer as an example ($l \leq L$), we denote the M-length word tokens as $T_l \in \mathbb{R}^{M \times C}$, which represent the input instruction and the already generated response. The learnable

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3 LLAMA-ADAPTER

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3.1 适配提示词

对于拥有 N层Transformer结构的预训练LLaMA模型，我们首先在其最顶层的 L 层（即 $L \leq N$层）中插入一组可学习适配提示。我们将这些提示记为 $\{P_l\}_{l=1}^{L}$，其中 $P_l \in \mathbb{R}^{K \times C}$与 K 分别表示每层的提示长度，而 C 则等于LLaMA Transformer的特征维度。在最后的 L 层使用这类提示词，能够更有效地调整语言表征，从而融入更高层级的语义信息。

以第 l个插入的层为例（即$l \leq L$），我们将长度为 M的词元记为$T_l \in \mathbb{R}^{M \times C}$，这些词元分别代表输入的指令以及已生成的响应。而那些可学习的参数

adaption prompt is concatenated with T_l along the token dimension as prefixes, formulated as

$$[P_l;\ T_l] \in \mathbb{R}^{(K+M)\times C}. \tag{1}$$

In this way, the instruction knowledge learned within P_l, can effectively guide T_l to generate the subsequent contextual response via our zero-initialized attention layers in the transformer block.

3.2 ZERO-INITIALIZED ATTENTION

If the adaption prompts are randomly initialized, they might bring disturbance to the word tokens at the beginning of training, which harms the fine-tuning stability and effectiveness. Considering this, we modify the vanilla self-attention at the last L layers to be zero-initialized variants, as shown in Figure 2. Suppose the model is generating the $(M+1)$-th word on top of $[P_l;\ T_l]$ at the l-th inserted layer, we denote the corresponding $(M+1)$-th word token as $t_l \in \mathbb{R}^{1\times C}$. In the attention mechanism, several linear projection layers are first applied to transform the input tokens into queries, keys, and values as

$$Q_l = \text{Linear}_\text{q}(\ t_l\ ); \tag{2}$$

$$K_l = \text{Linear}_\text{k}(\ [P_l;\ T_l;\ t_l]\ ); \tag{3}$$

$$V_l = \text{Linear}_\text{v}(\ [P_l;\ T_l;\ t_l]\ ). \tag{4}$$

Then, the attention scores of Q_l and K_l before the softmax function are calculated as

$$S_l = Q_l K_l^T/\sqrt{C} \in \mathbb{R}^{1\times(K+M+1)}, \tag{5}$$

which records the feature similarities between the new word t_l and all $K+M+1$ tokens. Meanwhile, S_l can be reformulated by two components as

$$S_l = [S_l^K;\ S_l^{M+1}]^T, \tag{6}$$

where $S_l^K \in \mathbb{R}^{K\times 1}$ and $S_l^{M+1} \in \mathbb{R}^{(M+1)\times 1}$ denote the attention scores of K adaption prompts and $M+1$ word tokens, respectively. The former S_l^K represents how much information the learnable prompt contributes to generating t_l, which probably causes disturbance in the early training stage.

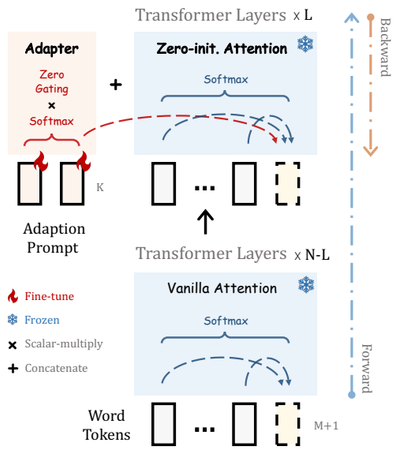


Figure 2: **Details of Zero-initialized Attention.** We insert learnable adaption prompts into the last L out of N transformer layers of LLaMA. To progressively learn the instructional knowledge, we adopt a zero gating factor within the attention for stable training in the early training stages.

To this end, we adopt a learnable gating factor, denoted as g_l, to adaptively control the importance of S_l^K in the attention. Initialized by zero, g_l can firstly eliminate the influence of under-fitted prompts, and then increase its magnitude for providing more instruction semantics to LLaMA. Therefore, we independently apply the softmax functions to the two components in Equation equation 6, and multiply the first term by g_l, formulated as

$$S_l^g = [\text{softmax}(S_l^K)\cdot\tanh(g_l);\ \text{softmax}(S_l^{M+1})]^T, \tag{7}$$

where an activation function $\tanh(\cdot)$ is adopted to regulate the scale of g_l to into -1∼1. The separate softmax functions ensure the second term to be irrelevant to the adaption prompts, and we do not multiply any coefficient to $\text{softmax}(S_l^{M+1})$ to prevent the pre-trained knowledge from being disturbed, i.e., preserving its original probability distribution. When g_l is close to zero, it can mostly convey the originally pre-trained knowledge of LLaMA to token t_l for a creditable generation. In practice, we adopt multiple g_l to be independently learned for different heads within the attention, benefiting the learning diversity of multi-head mechanisms.

Finally, we calculate the output of the l-th attention layer with a linear projection layer as

$$t_l^o = \text{Linear}_\text{o}(S_l^g V_l) \in \mathbb{R}^{1\times C}. \tag{8}$$

With our proposed zero-initialized attention, the adaption prompts can progressively inject the newly acquired instructional signals into the transformer, while simultaneously incorporating the pre-trained knowledge of LLaMA to provide high-quality responses.

该适配用的提示词会与 T_l 一起沿着标记维度作为前缀被连接起来，进而形成具体的表达形式。

$$[P_l;\ T_l] \in \mathbb{R}^{(K+M)\times C}. \tag{1}$$

In t通过这种方式， P_l中学习到的指令知识能够有效引导 T_l 生成内容。 e
sub通过变换器块中经过零初始化注意力机制处理的层层层次结构，生成连续的上下文相关响应。

3.2 零初始化注意力机制

If the adaption prompts are randomly initialized, they might bring disturbance to the word tokens at the beginning of training, which harms the fine-tuning stability and effectiveness. Considering this, we modify the vanilla self-attention at the last L layers to be zero-initialized variants, as shown in Figure 2. Suppose the model is generating the $(M+1)$-th word on top of $[P_l;\ T_l]$ at the l-th inserted layer, we denote the corresponding $(M+1)$-th word token as $t_l \in \mathbb{R}^{1\times C}$. In the attention mechanism, several linear projection layers are first applied to transform the input tokens into queries, keys, and values as

$$Q_l = \text{Linear}_\text{q}(\ t_l\ ); \tag{2}$$

$$K_l = \text{Linear}_\text{k}(\ [P_l;\ T_l;\ t_l]\ ); \tag{3}$$

$$V_l = \text{Linear}_\text{v}(\ [P_l;\ T_l;\ t_l]\ ). \tag{4}$$

接着，计算在应用Softmax函数之前 Q_l 和 K_l 的注意力分数。

$$S_l = Q_l K_l^T/\sqrt{C} \in \mathbb{R}^{1\times(K+M+1)}, \tag{5}$$

它记录了新词 t_l 与所有 $K+M+1$ 标记之间的特征相似度。同时， S_l 可以通过两个组成部分来重新表述

$$S_l = [S_l^K;\ S_l^{M+1}]^T, \tag{6}$$

其中 $S_l^K \in \mathbb{R}^{K\times 1}$ 和 $S_l^{M+1} \in \mathbb{R}^{(M+1)\times 1}$ 分别表示 K 适配提示词的注意力得分以及 $M+1$ 词元的注意力得分。 for-
mer S_l^K表示可学习的提示词在生成 t_l的过程中贡献了多少信息，而这很可能会在训练初期造成干扰。

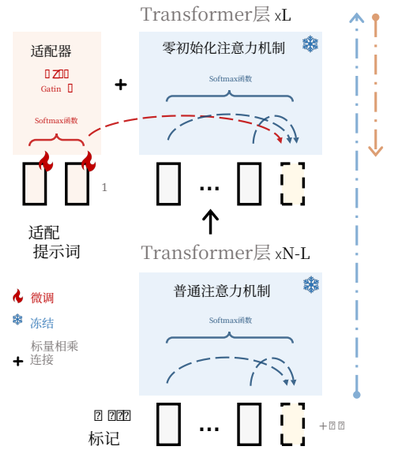


图2：**零初始化注意力机制的详细结构**。我们将可学习适配提示词插入到LLaMA的 N 个Transformer层中的最后L 层中。为了逐步学习指令相关知识，我们在注意力机制中采用了零门控机制，从而在训练初期实现稳定的训练过程。

为此，我们采用了一个可学习的门控因子，记为 g_l，用以自适应地控制S_l^K在注意力机制中的重要性。由于该因子初始值为零， g_l 它首先能够消除拟合度不足的提示词所带来的影响，随后再增大其数值，从而为LLaMA提供更丰富的指令语义信息。因此，我们在方程式6中的两个分量上分别应用Softmax函数，并将第一项乘以 g_l，具体表达形式为

$$S_l^g = [\text{softmax}(S_l^K)\cdot\tanh(g_l);\ \text{softmax}(S_l^{M+1})]^T, \tag{7}$$

其中采用了激活函数双曲正切函数(·)来调节 g_l 的数值范围，使其变为-1∼1。独立的Softmax函数可确保第二项与适配提示词无关，同时我们也不会对Softmax(S_l^{M+1})乘以任何系数，以避免破坏预训练知识，从而保持其原有的概率分布。当 g_l 接近零时，它便能主要传递LLaMA原有的预训练知识给标记 t_l ，进而实现可靠的文本生成。在实际应用中，我们会为注意力机制中的不同分支独立学习多个 g_l ，以此提升多头机制的学习多样性。

最后，我们通过线性投影层计算第 l个注意力层的输出值。

$$t_l^o = \text{Linear}_\text{o}(S_l^g V_l) \in \mathbb{R}^{1\times C}. \tag{8}$$

借助我们提出的零初始化注意力机制，适配提示词能够逐步将新获取的指令信号注入到Transformer模型中，同时结合LLaMA的预训练知识，从而生成高质量响应。

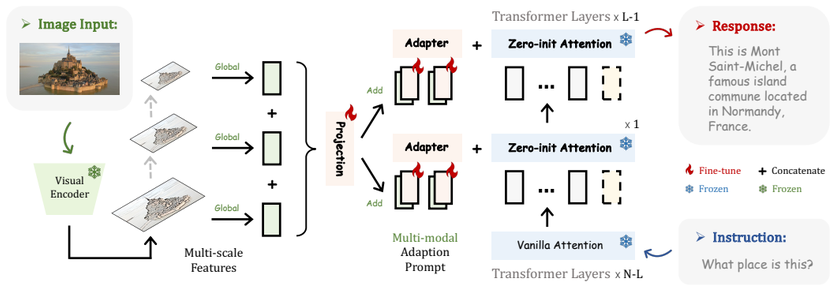


Figure 3: **Multi-modal LLaMA-Adapter.** By connecting a pre-trained image encoder, LLaMA-Adapter can be extended to a multi-modal LLM for image-conditioned instruction following. Given an image input, we element-wisely add the image tokens with adaption prompts, and utilize our zero-initialized attention mechanism to inject visual semantics into LLaMA (Touvron et al., 2023).

3.3 Multi-modal Reasoning

Apart from language instructions, LLaMA-Adapter is capable of answering a question based on image input with simple modifications. This fully unleashes the multi-modal reasoning power of LLMs for extensive application scenarios, e.g., image captioning, object counting, and OCR. The overall framework of our multi-modal LLaMA-Adapter is shown in Figure 3.

Multi-modal Architecture. For an input image, we first leverage a pre-trained visual encoder, e.g., CLIP (Radford et al., 2021), to extract its multi-scale global features, denoted as $\{I_m\}_{m=1}^{M}$, where $I_m \in \mathbb{R}^{1\times C_m}$ and M denotes the scale number. Then, we concatenate the M-scale features along the channel dimension, and apply a learnable projection network to transform them into word embedding space, formulated as

$$I_p = \text{Projection}\Big(\text{Concat}\big(\{I_m\}_{m=1}^{M}\big)\Big), \tag{9}$$

where $I_p \in \mathbb{R}^{1\times C}$ and is regarded as the overall image token with the same feature dimension as our adaption prompts. After this, we repeat I_p for K times, and element-wisely add it onto the K-length adaption prompts at all L inserted transformer layers. For the l-th layer, we denote the acquired multi-modal prompt as

$$P_l^v = P_l + \text{Repeat}(I_p) \ \in \mathbb{R}^{K\times C}, \tag{10}$$

where P_l^v denotes the prompt incorporating visual information from the given image. After this, our zero-initialized attention can learn to increasingly infuse the image-conditional semantics into LLaMA by the zero gating factor g_l. In this way, an LLM can be efficiently tuned to understand vision-language input, and tackle more challenging generative tasks with multi-modal reasoning.

Training Strategy. Instead of using Alpaca's data (Taori et al., 2023) for language-only instruction tuning, we fine-tune LLaMA-Adapter with multi-modal instruction data, and evaluate the performance with two popular scenarios:

- **ScienceQA (Lu et al., 2022) Evaluation.** ScienceQA includes a large-scale science question answering data collected from a wide range of knowledge domains. Each sample contains a visual context, a textual context, a question with multiple options, and an answer. We directly utilize ScienceQA's multi-modal training set to fine-tune LLaMA-Adapter, and conduct in-domain testing. We freeze both the image encoder and LLaMA, and only train the lightweight projection network and zero-initialized attention mechanisms.
- **Zero-shot Multi-modal Evaluation.** To verify the out-of-domain generation ability of our approach, we conduct a two-stage multi-modal training, and then evaluate three benchmarks (MME (Fu et al., 2023), MMBench (Liu et al., 2023c), LVLM-eHub (Xu et al., 2023))

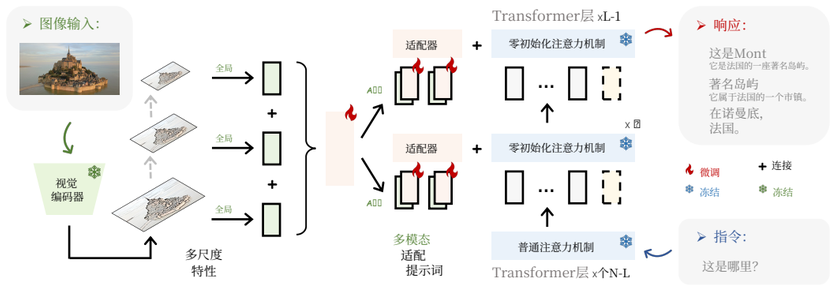


图3：**多模态LLaMA-Adapter**。通过连接预训练的图像编码器，LLaMA-Adapter能够被扩展为可用于图像条件指令遵循的多模态大语言模型。在接收到图像输入后，我们会针对每个元素将图像标记与适配提示词相结合，并利用我们经过零初始化的注意力机制将视觉语义注入到LLaMA中（Touvron等人，2023年）。

3.3 多模态推理

除了语言相关的指令之外，经过简单修改后，LLaMA-Adapter也能够基于图像输入来回答问题。这样一来，大语言模型的多模态推理能力便能得到充分释放，可应用于图像文本生成、物体计数以及光学字符识别等诸多场景。我们设计的这款多模态LLaMA-Adapter的整体框架如图3所示。

Multi-modal Architecture. For an input image, we first leverage a pre-trained visual encoder, e.g., CLIP (Radford et al., 2021), to extract its multi-scale global features, denoted as $\{I_m\}_{m=1}^{M}$, where $I_m \in \mathbb{R}^{1\times C_m}$ and M denotes the scale number. Then, we concatenate the M-scale features along the channel dimension, and apply a learnable projection network to transform them into word embedding space, formulated as

$$I_p = \text{Projection}\Big(\text{Concat}\big(\{I_m\}_{m=1}^{M}\big)\Big), \tag{9}$$

其中 $I_p \in \mathbb{R}^{1\times C}$ 被视为具有与我们的适配提示词相同特征维度的整体图像令牌。之后，我们重复 I_p 操作 K 次，并在所有 L 个插入的Transformer层中，将结果逐元素地添加到长度为 K的适配提示词上。对于第 l层，我们将获取到的多模态提示词记为

$$P_l^v = P_l + \text{Repeat}(I_p) \ \in \mathbb{R}^{K\times C}, \tag{10}$$

其中 P_l^v表示融入了给定图像中视觉信息的提示词。此后，通过零门控因子 g_l，我们采用零初始化的注意力机制逐渐将基于图像的条件语义注入LLaMA中。如此一来，大语言模型就能被高效地调优，从而理解视觉语言输入，并借助多模态推理来完成更具挑战性的生成任务。

训练策略。我们没有使用Alpaca的数据（Taori等人，2023年）来进行仅基于文本的指令微调，而是采用多模态指令数据对LLaMA-Adapter进行微调，并通过两种常见的应用场景来评估其性能：

- **ScienceQA（卢等人，2022年）评估**。ScienceQA包含从众多知识领域收集而来的大规模科学问答数据集。每个样本都包含视觉上下文、文本上下文、带有多个选项的题目以及答案。我们直接利用ScienceQA的多模态训练集对LLaMA-Adapter进行微调，并开展领域内测试。在此过程中，我们会冻结图像编码器和LLaMA模型本身，仅对轻量级的投影网络以及零初始化注意力机制进行训练。
- **零样本多模态评估**。为验证该方法在跨领域生成任务上的能力，我们首先进行了两阶段的多模态训练，随后在三项基准测试上进行了评估：MME（Fu等人，2023年）、MMBench（Liu等人，2023年c版）以及LVLM-eHub（Xu等人，2023年）。

Instruction: Tell me about alpacas.

Alpaca: Alpacas are small, fluffy animals related to camels and llamas. They are native to Peru and Bolivia, and were first domesticated around 5,000 years ago. They are kept mainly for their fine, soft fleece, which is used to make knitwear and other garments. Alpacas are herd animals and live in small family groups. They are highly social animals and can form strong bonds with their owners.

LLaMA-Adapter: Alpacas are members of the camelid family and are native to the Andes Mountains of South America. They are typically found in herds of 10-20 animals and are known for their gentle nature and their ability to thrive in harsh climates. Alpacas are typically white or grey in color and have long, silky coats. They are also known for their ability to produce high-quality wool, which is used in the production of clothing and other products.

Instruction: Translate the sentence 'I have no mouth but I must scream' into Spanish.

Alpaca: No tengo boca, pero debo gritar.

LLaMA-Adapter: No tengo boca pero tengo que gritar.

Instruction: Write a Python program that prints the first 10 Fibonacci numbers.

Alpaca:

```
numbers = [0, 1]
for i in range(2, 11):
    numbers.append(numbers[i-2] + numbers[i-1])
print(numbers)
```

LLaMA-Adapter:

```
def print_fibonacci(n):
    a, b = 0, 1
    count = 0
    while count < n:
        print(a, end=' ')
        a, b = b, a + b
        count += 1
print_fibonacci(10)
```

Figure 4: **Language Instruction-following Capacity.** Our LLaMA-Adapter performs comparably with Alpaca (Taori et al., 2023) for question answering, language translation, and code generation.

in a zero-shot manner. For the first stage, we utilize the raw image-caption data from LAION-400M (Schuhmann et al., 2021) to tune the projection network and zero-initialized attention modules. This step is mainly for the alignment of embedding space between visual features and word tokens. For the second stage, we freeze the projection network, and only tune the zero-initialized attention within LLaMA by a combination of Alpaca's data and LLaVA-I (Liu et al., 2023b). The second stage aims to endow LLMs with the capability to generate detailed responses based on human instructions.

4 EXPERIMENT

In Section 4.1, we first evaluate the language instruction-following capacity of LLaMA-Adapter. Then, we present our multi-modal reasoning performance on several benchmarks in Section 4.2, and conduct ablation studies on ScienceQA's validation set in Section 4.3. Finally, we report the fine-tuning results of our approach on traditional vision and language models in Section 4.4.

4.1 INSTRUCTION-FOLLOWING EVALUATION

Settings. Following Stanford Alpaca (Taori et al., 2023), we utilize 52K instruction-following data for training. We fine-tune LLaMA-Adapter on 8 A100 GPUs for 5 epochs. The warmup epochs, batch size, learning rate, and weight decay are set to 2, 64, 0.009, and 0.02, respectively. By default, we utilize the pre-trained LLaMA model with 7B parameters and $N = 32$ transformer layers. We

指令：请为我介绍一下羊驼。

Alpaca：羊驼是一种体型较小、毛茸茸的动物，与骆驼和羊驼属动物有亲缘关系。它们原产于秘鲁以及玻利维亚，大约在5000年前开始被人类驯养。人们饲养它们主要是为了获取其细腻柔软的毛皮。羊驼毛被用于制作针织品及其他服饰。羊驼属于群居动物，通常以小群体形式生活。它们是极具社交性的动物，能够与主人建立起深厚的感情纽带。

LLaMA-Adapter: 羊驼隶属于骆驼科，原产于南美洲的安第斯山脉。它们一般以10到20只为一群活动，以其温顺的性格以及在恶劣气候条件下仍能良好生存的能力而闻名。羊驼的颜色多为白色或灰色，而且拥有长而柔顺的毛皮。此外，它们还以能够产出高品质羊毛而著称，这种羊毛品质上乘。被用于服装及其他产品的生产中。

指令：将句子‘I have no mouth but I must scream’翻译成西班牙语。

Alpaca模型：No tengo boca, pero debo gritar.

LLaMA-Adapter：我虽无嘴，却不得不呼喊。

指令：编写一个Python程序，输出前10个斐波那契数列的数值。

Alpaca模型：

```
数字:  = [0  , 1]
用于变量 i在range(2,   11):
    numbers.append(numbers[+i-2] +  +numbers[+i-1])
print(numbers)
```

LLaMA-Adapter:

```
def print_fibonacci(n):
    a, b = 0, 1
    计数 = 0
    while计数 < n:
        打印(a.      结束=' ')
        a, b =b , a + b
        计数 += 1
打印_斐波那契数列(10)的结果
```

图4：**语言指令遵循能力**。在问答、语言翻译以及代码生成任务中，我们的LLaMA-Adapter表现与Alpaca模型（Taori等人，2023年）相当。

以零样本方式实现。在第一阶段，我们利用来自LAION-400M的原始图像-文本生成数据（Schuhmann等人，2021年）来调整投影网络以及零初始化的注意力模块。此步骤的主要目的是使视觉特征与词元之间的嵌入空间保持一致。在第二阶段，我们会冻结投影网络，仅通过结合Alpaca模型与LLaVA-I（Liu等人，2023b年）的数据来调整LLaMA内部的零初始化注意力机制。第二阶段的目标是让大语言模型具备根据人类指令生成详细响应的能力。

4 实验

在4.1节中，我们首先评估LLaMA-Adapter的语言指令遵循能力。接着，在4.2节里，我们会展示该ALL模型在多个基准测试中的多模态推理表现，并在4.3节对ScienceQA数据集的验证集开展消融实验。最后，在4.4节，我们将汇报该方法在传统视觉与语言模型上的微调结果。

4.1 指令遵循能力评估

设置选项。参照Stanford Alpaca（Taori等人，2023年）的研究，我们使用了52K条指令遵循数据来进行训练。我们在8块A100 GPU上对LLaMA-Adapter进行微调，共经历5个训练轮数。预热轮数、批量大小、学习率以及权重衰减值分别被设置为2、64、0.009和0.02。默认情况下，我们使用的是具有70亿参数的预训练LLaMA模型以及 $N = 32$Transformer层。我们

Figure 5: **GPT-4 Evaluating Benchmark (Chiang et al., 2023)** for LLaMA-Adapter, Alpaca and Alpaca-LoRA.

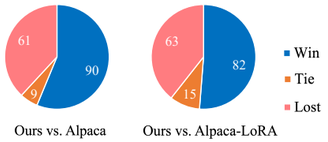


Table 1: **Efficiency Comparison.** The training time is tested on 8 A100 GPUs.

Model	Tuned Params	Storage Space	Training Time
Alpaca	7B	13G	3 hours
Alpaca-LoRA	4.2M	16.8M	1.5 hours
LLaMA-Adapter	**1.2M**	**4.7M**	**1 hour**

Table 2: **Visual Question Answering on ScienceQA (Lu et al., 2022) Benchmark.** CoT denotes using a chain of thought for question answering. T denotes using text-only input.

Model	Tuned Params	Avg	NAT	SOC	LAN	TXT	IMG	NO	G1-6	G7-12
Random Choice (Lu et al., 2022)	-	39.83	40.28	46.13	29.25	47.45	40.08	33.66	39.35	40.67
Human (Lu et al., 2022)	-	88.40	90.23	84.97	87.48	89.60	87.50	88.10	91.59	82.42
ChatGPT$_{CoT}$ (OpenAI, 2023a)	0M	78.31	78.82	70.98	83.18	77.37	67.92	86.13	80.72	74.03
GPT-4$_{CoT}$ (OpenAI, 2023b)	0M	83.99	85.48	72.44	90.27	82.65	71.49	92.89	86.66	79.04
MCAN (Yu et al., 2019)	95M	54.54	56.08	46.23	58.09	59.43	51.17	55.40	51.65	59.72
VisualBERT (Li et al., 2019a; 2020)	111M	61.87	59.33	69.18	61.18	62.71	62.17	58.54	62.96	59.92
UnifiedQA (Khashabi et al., 2020)	223M	70.12	68.16	69.18	74.91	63.78	61.38	77.84	72.98	65.00
UnifiedQA$_{CoT}$	223M	74.11	71.00	76.04	78.91	66.42	66.53	81.81	77.06	68.82
MM-COT$_T$ (Zhang et al., 2023e)	223M	70.53	71.09	70.75	69.18	71.16	65.84	71.57	71.00	69.68
MM-COT	223M	84.91	87.52	77.17	85.82	87.88	82.90	86.83	84.65	85.37
LLaMA-Adapter$_T$	1.2M	78.31	79.00	73.79	80.55	78.30	70.35	83.14	79.77	75.68
LLaMA-Adapter	1.8M	85.19	84.37	88.30	84.36	83.72	80.32	86.90	85.83	84.05

adopt a prompt length $K = 10$ and insert the adaption prompts into the last $L = 30$ layers. For quantitative results, we compare with methods both trained by the 52K instruction data, Alpaca (Taori et al., 2023) and Alpaca-LoRA (alp, 2023), and evaluate with one widely adopted schemes, GPT-4 evaluating benchmark (Chiang et al., 2023). It adopts GPT-4 (OpenAI, 2023b) to assess the quality of two compared responses from different models on 80 questions.

Performance. We first show some generated responses of LLaMA-Adapter and Alpaca in Figure 4. For different kinds of instructions, our approach can output reasonable responses comparable to the fully fine-tuned Alpaca, including question answering, language translation, and code generation. Please refer to the Appendix for a full comparison with Alpaca-LoRA, GPT-3 (Brown et al., 2020), and LLaMA-I (Touvron et al., 2023). For GPT-4 assessment in Figure 5, LLaMA-Adapter obtains more 'win' compared to Alpaca and Alpaca-LoRA, respectively. This fully demonstrates the effectiveness of our adaption method with zero-initialized attention mechanisms.

Efficiency. In Table 1, we compare the learnable parameters, storage space, and training time of different instruction-following methods. As a lightweight plug-and-play module, LLaMA-Adapter enjoys superior training efficiency with only 1.2M parameters, 4.9M storage, and one-hour training. This enables more efficient storage of large-scale language models on mobile devices. LLaMA-Adapter's efficiency advantages can be further revealed by multi-node training, since only the gradients of 1.2M parameters are required to be transferred among nodes, other than Alpaca's 7B.

4.2 Multi-modal Evaluation

Settings. We adopt CLIP (Radford et al., 2021) as the image encoder to extract multi-scale visual features, and leverage a simple bottleneck MLP layer as the learnable projection network. We keep other hyperparameters the same as the language instruction-following LLaMA-Adapter. For ScienceQA (Lu et al., 2022), we concatenate the given question, textual context, and options sequentially in one sentence as LLaMA's input. For zero-shot multi-modal evaluation, we select three benchmarks, MME (Fu et al., 2023), MMBench (Liu et al., 2023c), and LVLM-eHub (Xu et al., 2023), covering a wide range of VQA tasks. We compare with two concurrent multi-modal LLMs: LLaVA (Liu et al., 2023b) and MiniGPT-4 (Zhu et al., 2023).

图5：**GPT-4评估基准（Chiang等人，2023年）**，用于测试LLaMA-Adapter、Alpaca模型以及Alpaca-LoRA。

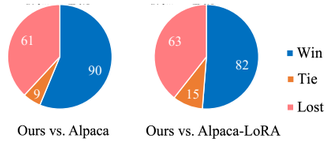


表1：**效率对比**。训练时间是在8台A100 GPU上测试得出的。

模型	经过调优的模型 参数量	存储空间 空间	训练 Time
Alpaca模型	7B	13G	3小时
Alpaca-LoRA	4.2M	1680万	1.5小时
LLaMA-Adapter	**1.2M**	**4.7M**	**1小时**

表2：**基于ScienceQA数据集的视觉问答任务（卢等人，2022年）测试结果**。 CoT 表示在问答过程中使用了思维链技术。 T 表示仅使用文本输入进行测试。

模型	经过微调的模型 参数量	Avg	NAT	SOC	LAN	TXT	IMG	NO	G1-6	G7-12
随机选择方法（卢等人、2022年）	-	39.83	40.28	46.13	29.25	47.45	40.08	33.66	39.35	40.67
人类参与者（卢等人，2022年）	-	88.40	90.23	84.97	87.48	89.60	87.50	88.10	91.59	82.42
ChatGPT$_{CoT}$ (OpenAI,2023a年)	0M	78.31	78.82	70.98	83.18	77.37	67.92	86.13	80.72	74.03
GPT-4$_{CoT}$ (OpenAI、2023b)	0M	83.99	85.48	72.44	90.27	82.65	71.49	92.89	86.66	79.04
MCAN方法（余等人，2019年）	95M	54.54	56.08	46.23	58.09	59.43	51.17	55.40	51.65	59.72
VisualBERT（李等人，2019a年；2020年）	111M	61.87	59.33	69.18	61.18	62.71	62.17	58.54	62.96	59.92
UnifiedQA（Khashabi等人，2020年）	223M	70.12	68.16	69.18	74.91	63.78	61.38	77.84	72.98	65.00
UnifiedQA$_{CoT}$	223M	74.11	71.00	76.04	78.91	66.42	66.53	81.81	77.06	68.82
MM-COT$_T$(Zhang等人，2023e年)	223M	70.53	71.09	70.75	69.18	71.16	65.84	71.57	71.00	69.68
MM-COT	223M	84.91	87.52	77.17	85.82	87.88	82.90	86.83	84.65	85.37
LLaMA-Adapter$_T$	1.2M	78.31	79.00	73.79	80.55	78.30	70.35	83.14	79.77	75.68
LLaMA-Adapter	1.8M	85.19	84.37	88.30	84.36	83.72	80.32	86.90	85.83	84.05

adopt a prompt length $K = 10$ and insert the adaption prompts into the last $L = 30$ layers. For quantitative results, we compare with methods both trained by the 52K instruction data, Alpaca (Taori et al., 2023) and Alpaca-LoRA (alp, 2023), and evaluate with one widely adopted schemes, GPT-4 evaluating benchmark (Chiang et al., 2023). It adopts GPT-4 (OpenAI, 2023b) to assess the quality of two compared responses from different models on 80 questions.

Performance. We first show some generated responses of LLaMA-Adapter and Alpaca in Figure 4. For different kinds of instructions, our approach can output reasonable responses comparable to the fully fine-tuned Alpaca, including question answering, language translation, and code generation. Please refer to the Appendix for a full comparison with Alpaca-LoRA, GPT-3 (Brown et al., 2020), and LLaMA-I (Touvron et al., 2023). For GPT-4 assessment in Figure 5, LLaMA-Adapter obtains more 'win' compared to Alpaca and Alpaca-LoRA, respectively. This fully demonstrates the effectiveness of our adaption method with zero-initialized attention mechanisms.

效率。在表1中，我们对比了不同指令遵循方法的可学习参数、存储空间以及训练时间。作为一款轻量级且即插即用的模块，LLaMA-Adapter仅需120万参数、490万存储空间，加上1小时的训练时间，就能展现出极高的训练效率。这为在移动设备上更高效地存储大规模语言模型提供了可能。通过多节点训练还能进一步凸显LLaMA-Adapter的效率优势，因为与需要传输70亿参数梯度的Alpaca模型不同，它仅需在各个节点之间传递120万参数的梯度即可。

4.2 多模态评估

Settings. We adopt CLIP (Radford et al., 2021) as the image encoder to extract multi-scale visual features, and leverage a simple bottleneck MLP layer as the learnable projection network. We keep other hyperparameters the same as the language instruction-following LLaMA-Adapter. For ScienceQA (Lu et al., 2022), we concatenate the given question, textual context, and options sequentially in one sentence as LLaMA's input. For zero-shot multi-modal evaluation, we select three benchmarks, MME (Fu et al., 2023), MMBench (Liu et al., 2023c), and LVLM-eHub (Xu et al., 2023), covering a wide range of VQA tasks. We compare with two concurrent multi-modal LLMs: LLaVA (Liu et al., 2023b) and MiniGPT-4 (Zhu et al., 2023).

Table 3: **Zero-shot Multi-modal Evaluation** on MME (Fu et al., 2023), MMBench (Liu et al., 2023c) and LVLM-eHub (Xu et al., 2023) benchmarks. P: Perception; C: Cognition. LR: Logical Reasoning; AR: Attribute Reasoning; RR: Relation Reasoning; FP-C/S: Fine-grained Perception (Cross Instance/Single Instance); CP: Coarse Perception. VP: Visual Perception; VKA: Visual Knowledge Acquisition; VR: Visual Reasoning; VC: Visual Commonsense.

Model	MME		MMbench							LVLM-eHub			
	P	C	All	LR	AR	RR	FP-S	FP-C	CP	VP	VKA	VR	VC
LLaVA	503	215	36.2	15.9	53.6	28.6	41.8	20.0	40.4	0.62	0.38	0.77	0.79
Mini-GPT4	867	292	23.0	13.6	32.9	8.9	28.7	11.2	28.3	0.73	0.35	0.53	0.57
LLaMA-Adapter	973	249	39.5	13.1	47.4	23.0	45.0	33.2	50.6	0.81	0.44	0.83	0.59

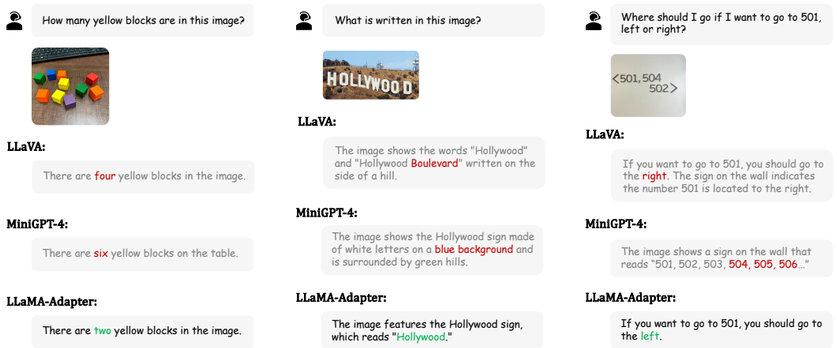


Figure 6: **Multi-modal Reasoning Examples** on zero-shot open-domain questions. We compare our approach with LLaVA (Liu et al., 2023b) and MiniGPT-4 (Zhu et al., 2023) for object counting, OCR, and common sense reasoning.

Performance. In Table 2 for the ScienceQA performance, our single-modal 'LLaMA-Adapter$_T$' attains 78.31% accuracy, surpassing several traditional VQA methods with large parameters. By further injecting visual conditions with a 0.6M projection network, our multi-modal 'LLaMA-Adapter' improves +6.88% accuracy, attaining leading results superior to the GPT series. In Table 3 for the three multi-modal benchmarks, compared to the concurrent works, our approach achieves competitive scores with a much more efficient tuning strategy. This is because, LLaVA requires fine-tuning the entire 7B LLM, and Mini-GPT4 adopts Vicuna (Chiang et al., 2023) that also fully fine-tunes LLaMA with 13B parameters. We also show some multi-modal reasoning examples in Figure 6. Our approach exhibits better object counting, OCR, and commence reasoning performance.

4.3 Ablation Study

Insertion Layers. We first investigate the number of transformer layers to be inserted by zero-initialized attention in LLaMA-Adapter. As shown in Table 4, increasing the layer numbers introduces more parameters, but leads to a large improvement in the answering accuracy of ScienceQA's validation set. There also exists an optimal insertion number from the higher layers, since too many layers would adversely disturb the early encoding of input words. If one has limited resources to identify the best number, simply inserting into all transformer layers is generally a good solution.

Zero-initialized Attention. Our proposed zero-initialized attention is essential for the early-stage training stability and final generation capacity. As shown in Table 5, it contributes to a significant +43.08% gain on ScienceQA's validation set. In contrast, the randomly initialized baseline only achieves 40.77% accuracy, nearly the same as 'Random Choice' (Table 2's first row). In Figure 7, we plot the loss curves with and without the zero initialization. The loss of 'Zero-initialized' declines much faster at the beginning, and finally converges to zero. In contrast, the 'Random-initialized' slowly approaches 0.15, which is not fully converged and causes a large performance drop.

Table 3: **Zero-shot Multi-modal Evaluation** on MME (Fu et al., 2023), MMBench (Liu et al., 2023c) and LVLM-eHub (Xu et al., 2023) benchmarks. P: Perception; C: Cognition. LR: Logical Reasoning; AR: Attribute Reasoning; RR: Relation Reasoning; FP-C/S: Fine-grained Perception (Cross Instance/Single Instance); CP: Coarse Perception. VP: Visual Perception; VKA: Visual Knowledge Acquisition; VR: Visual Reasoning; VC: Visual Commonsense.

模型	MME		MMbench基准测试							LVLM-eHub基准测试			
	P	C	All	LR	AR	RR	假阳性率 假阴性率		CP	VP	VKA	VR	VC
LLaVA	503	215	36.2	15.9	53.6	28.6	41.8	20.0	40.4	0.62	0.38	0.77	0.79
Mini-GPT4	867	292	23.0	13.6	32.9	8.9	28.7	11.2	28.3	0.73	0.35	0.53	0.57
LLaMA-Adapter	973	249	39.5	13.1	47.4	23.0	45.0	33.2	50.6	0.81	0.44	0.83	0.59

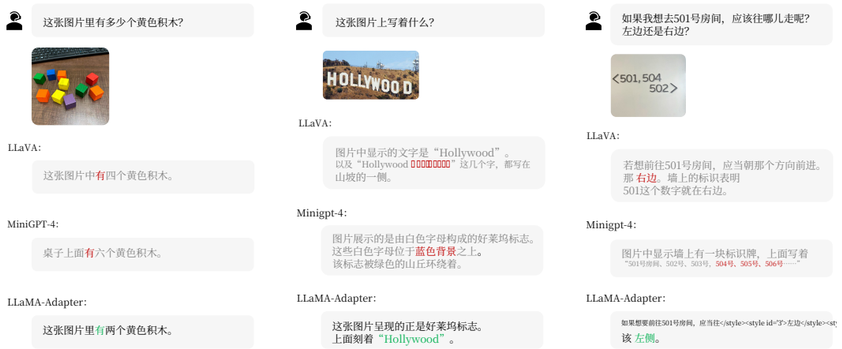


图6：**多模态推理示例**在零样本开放领域问题上的应用。我们将本方法与LLaVA（Liu等人，2023b年）以及MiniGPT-4（Zhu等人，2023年）在物体计数、光学字符识别以及常识推理任务上的表现进行了对比。

Performance. In Table 2 for the ScienceQA performance, our single-modal 'LLaMA-Adapter$_T$' attains 78.31% accuracy, surpassing several traditional VQA methods with large parameters. By further injecting visual conditions with a 0.6M projection network, our multi-modal 'LLaMA-Adapter' improves +6.88% accuracy, attaining leading results superior to the GPT series. In Table 3 for the three multi-modal benchmarks, compared to the concurrent works, our approach achieves competitive scores with a much more efficient tuning strategy. This is because, LLaVA requires fine-tuning the entire 7B LLM, and Mini-GPT4 adopts Vicuna (Chiang et al., 2023) that also fully fine-tunes LLaMA with 13B parameters. We also show some multi-modal reasoning examples in Figure 6. Our approach exhibits better object counting, OCR, and commence reasoning performance.

4.3 ABLATION STUDY

插入层。首先，我们研究了在LLaMA-Adapter中通过零初始化注意力机制插入的Transformer层数量。如表4所示，增加层数虽然会引入更多参数，但能显著提升ScienceQA数据集验证集的回答准确率。此外，从较高层开始插入也存在一个最佳数量，因为层数过多反而可能干扰对输入词汇的早期编码。如果缺乏资源来确定最优数量，直接在所有Transformer层中插入通常也是一个不错的解决方案。

Zero-initialized Attention. Our proposed zero-initialized attention is essential for the early-stage training stability and final generation capacity. As shown in Table 5, it contributes to a significant +43.08% gain on ScienceQA's validation set. In contrast, the randomly initialized baseline only achieves 40.77% accuracy, nearly the same as 'Random Choice' (Table 2's first row). In Figure 7, we plot the loss curves with and without the zero initialization. The loss of 'Zero-initialized' declines much faster at the beginning, and finally converges to zero. In contrast, the 'Random-initialized' slowly approaches 0.15, which is not fully converged and causes a large performance drop.

Table 4: **Number of Insertion Layers** to the pre-trained transformer of LLaMA.

Layers	Params	Val Acc.
10	0.97	55.95
20	1.37	73.36
30	1.79	**83.85**
32	1.83	81.03

Table 5: Effectiveness of **Zero-initialized Attention** in our method.

Setting	Val Acc.
Rand-Init.	40.77
Zero-Init.	**83.85**
Gain	+43.08

Figure 7: **Loss Curves** of LLaMA-Adapter with (blue) and without (orange) zero-initialized attention.

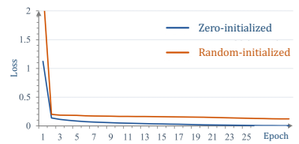


Table 6: **Vision Model Fine-tuning** with ViT on VTAB-1k benchmark.

Method	Natural	Special.	Struct.
Full	75.88	83.36	47.64
Adapter	70.39	77.11	33.43
Sidetune	58.21	68.12	23.41
VPT	78.48	82.43	54.98
Zero-init.	**81.74**	**84.43**	**56.75**

Table 7: **Language Model Fine-tuning** with RoBERTa on SQuAD benchmark.

Method	SQuAD 1.1		SQuAD 2.0	
	EM	F1	EM	F1
Full	88.9	94.6	86.5	89.4
PT	1.2	12.0	50.2	50.2
PT2	88.5	94.4	82.1	85.5
Zero-init.	**88.8**	**94.6**	**83.9**	**87.2**

Table 8: **Vision-language Fine-tuning** with CLIP on base-to-novel benchmark.

Method	Base	Novel	HM
CLIP	75.88	83.36	47.64
CoOp	70.39	77.11	33.43
CoCop	58.21	68.12	23.41
MaPLe	78.48	82.43	54.98
Zero-init.	**81.74**	**84.43**	**56.75**

4.4 Zero-initialized Attention for other Large Models

Our approach, i.e., zero-initialized attention, is not limited to the domain of tuning instruction models, and can be further utilized to fine-tune large models in traditional vision and language tasks.

Vision Models. We select a pre-trained ViT/16 (Dosovitskiy et al., 2020) as the vision model and evaluate on VTAB-1k (Zhai et al., 2019) benchmark, which contains 19 visual tasks with three domains: Natural, Specialized, and Structured. As shown in Table 6, for various image distributions, e.g., natural images, medical and satellite imagery, our approach performs much better than the full fine-tuning, and also surpasses existing parameter-efficient methods (Jia et al., 2022; Houlsby et al., 2019; Zhang et al., 2020), indicating our generalization ability for vision tasks.

Language Models. We utilize a pre-trained RoBERTa$_{\text{large}}$ (Liu et al., 2019) and adopt SQuAD (Rajpurkar et al., 2016) v1.1 and v2.0 benchmarks for extractive question answering evaluation. Exact Match (EM) and F1 scores on the dev set are reported. We refer to the Appendix for other language tasks. As shown in Table 7, the leading results among previous methods (Lester et al., 2021; Liu et al., 2021a) demonstrate our superiority over traditional language tasks.

Vision-language Models. We adopt CLIP (Radford et al., 2021) as the pre-trained vision-language model, and test on base-to-novel generalization (Zhou et al., 2022b) benchmark, where 'HM' denotes harmonic mean. As shown in Table 8, compared to previous works (Zhou et al., 2022a;c; Khattak et al., 2022), our approach achieves the best average classification accuracy on both base and novel categories, demonstrating our fine-tuning capability for large vision-language models.

5 Conclusion

In this paper, we propose LLaMA-Adapter, an efficient adaption method for tuning instruction-following models. For better training stability and final performance, we introduce the zero-initialized attention mechanism with a learnable gating factor, which increasingly incorporates instructional signals, while preserving the pre-trained knowledge in LLaMA. With only 1.2M parameters and one-hour training, our approach effectively fine-tunes LLaMA with superior efficiency compared to the 7B-parameter Alpaca. LLaMA-Adapter can be generalized to image-conditioned generation as a multi-modal LLM, achieving competitive results on various visual question answering benchmarks. On traditional vision and language tasks, our zero-initialized attention also attains favorable fine-tuning performance, which indicates strong generalization capacity.

表4：**LLaMA的预训练变换器所对应的插入层数量。**

层数	参数量	验证准确率
10	0.97	55.95
20	1.37	73.36
30	1.79	**83.85**
32	1.83	81.03

表5：本方法中**零初始化的自回归**机制的有效性。

设置选项	验证准确率
随机初始化	40.77
零初始化。	**83.85**
Gain	+43.08

图7：**损失曲线**，展示了采用零初始化注意力机制与未采用该机制的LLaMA-Adapter模型的对比情况（蓝色线条代表前者，橙色线条代表后者）。

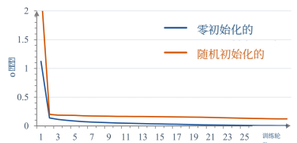


表6：**视觉模型微调**，展示了在VTAB-1k基准测试集上使用ViT模型进行微调的相关结果。

方法	自然语言	特别说明。	结构说明。
Full	75.88	83.36	47.64
适配器	70.39	77.11	33.43
旁路微调。	58.21	68.12	23.41
VPT	78.48	82.43	54.98
零初始化。	**81.74**	**84.43**	**56.75**

表7：**在SQuAD基准测试上使用RoBERTa模型进行语言模型微调。**

方法	SQuAD 1.1基准测试		SQuAD 2.0基准测试	
	EM	F1	EM	F1
Full	88.9	94.6	86.5	89.4
PT	1.2	12.0	50.2	50.2
PT2	88.5	94.4	82.1	85.5
零初始化。	**88.8**	**94.6**	**83.9**	**87.2**

表8：**在基础到新颖文本基准测试上使用CLIP模型进行的视觉语言微调。**

方法	Base	新颖的	HM
CLIP	75.88	83.36	47.64
CoOp	70.39	77.11	33.43
CoCop	58.21	68.12	23.41
MaPLe	78.48	82.43	54.98
零初始化。	**81.74**	**84.43**	**56.75**

4.4 其他大型模型的零初始化注意力机制

我们所采用的方法，也就是零初始化的注意力机制，并不仅限于指令模型的微调领域，还可以进一步用于传统视觉与语言任务中的大型模型微调。

Vision Models. We select a pre-trained ViT/16 (Dosovitskiy et al., 2020) as the vision model and evaluate on VTAB-1k (Zhai et al., 2019) benchmark, which contains 19 visual tasks with three domains: Natural, Specialized, and Structured. As shown in Table 6, for various image distributions, e.g., natural images, medical and satellite imagery, our approach performs much better than the full fine-tuning, and also surpasses existing parameter-efficient methods (Jia et al., 2022; Houlsby et al., 2019; Zhang et al., 2020), indicating our generalization ability for vision tasks.

Language Models. We utilize a pre-trained RoBERTa$_{\text{large}}$ (Liu et al., 2019) and adopt SQuAD (Rajpurkar et al., 2016) v1.1 and v2.0 benchmarks for extractive question answering evaluation. Exact Match (EM) and F1 scores on the dev set are reported. We refer to the Appendix for other language tasks. As shown in Table 7, the leading results among previous methods (Lester et al., 2021; Liu et al., 2021a) demonstrate our superiority over traditional language tasks.

Vision-language Models. We adopt CLIP (Radford et al., 2021) as the pre-trained vision-language model, and test on base-to-novel generalization (Zhou et al., 2022b) benchmark, where 'HM' denotes harmonic mean. As shown in Table 8, compared to previous works (Zhou et al., 2022a;c; Khattak et al., 2022), our approach achieves the best average classification accuracy on both base and novel categories, demonstrating our fine-tuning capability for large vision-language models.

5 结论

在本文中，我们提出了LLaMA-Adapter，这是一种用于微调指令遵循型模型的高效适配方法。为提升训练稳定性与最终性能，我们引入了带有可学习门控因子的零初始化注意力机制，该机制能够在保留LLaMA预训练知识的同时，逐步融入更多指令信号。仅需120万参数且仅需一小时训练时间，我们的方法就能以远超70亿参数版Alpaca模型的效率对LLaMA进行有效微调。LLaMA-Adapter还可被拓展应用于基于图像的条件生成场景，作为多模态大语言模型在各类视觉问答基准测试中取得优异成绩。在传统的视觉与语言任务上，这种零初始化注意力机制也同样展现出出色的微调效果，进一步证明了其强大的泛化能力。

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6 致谢

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A Overview

- Section B: Detailed results of zero-shot multi-modal evaluation.
- Section C: Additional related work.
- Section D: Detailed results of fine-tuning traditional vision and language models.
- Section E: Additional experiments and discussion.
- Section F: Full comparison of instruction-following models.
- Section G: Comparison of LLaMA-Adapter and LLaMA-I.

B More Details of Multi-modal Evaluation

ScienceQA (Lu et al., 2022) Evaluation. The data sample in ScienceQA contains a visual context, a textual context, a question, multiple options, and a correct answer, as shown in Figure 8. We omit the lecture and explanation in some data samples for simplicity.

Zero-shot Multi-modal Evaluation. We test our approach on the three benchmarks (Fu et al., 2023; Liu et al., 2023c; Xu et al., 2023) following their official procedures. In Tables 9, 10 and 11, we respectively report the detailed results of MME and LVLM-eHub benchmarks. As shown, across a wide range of visual question-answering problems, our approach can consistently achieve competitive results. We also show more examples of the multi-modal LLaMA-Adapter for open-domain zero-shot visual questions in Figure 9, where our approach can generate detailed and high-quality responses in natural language. The experiments fully demonstrate the generalization capacity of our proposed multi-modal LLM. We also give some qualitative examples in Figures 9 and 10, where our LLaMA-Adapter can answer open-ended questions for web images.

Table 9: **Perception Results on MME Benchmark (Fu et al., 2023).**

Model	ALL	Existence	Count	Position	Color	Poster	Celebrity	Scene	Landmark	Artwork	OCR
LLaVA	503	50	50	50	55	50	49	50	50	49	50
MiniGPT-4	867	115	123	82	110	56	65	96	69	56	83
LLaMA-Adapter	973	120	50	48	75	100	86	149	150	70	125

Table 10: **Cognition Results on MME Benchmark (Fu et al., 2023).**

Model	ALL	Commonsense Reasoning	Numerical Calculation	Text Translation	Code Reasoning
LLaVA	215	57	50	58	50
MiniGPT-4	292	72	55	55	110
LLaMA-Adapter	249	81	63	50	55

Table 11: **Zero-shot Multi-modal Results on LVLM-eHub Benchmark (Xu et al., 2023).** OC: Object Counting; MCI: Multi-Class Identification; KIE: Key Information Extraction; VE: Visual Entailment; KGID: Knowledge-grounded Image Description; VCR: Visual Commonsense Reasoning.

LVLM-eHub	Tasks	#Datasets	LLaVA	MiniGPT-4	LLaMA-Adapter
Visual Perception	ImgCls, OC, MCI	8	0.62	0.73	0.81
Visual Knowledge Acquisition	OCR, KIE, Caption	17	0.38	0.35	0.44
Visual Reasoning	VQA, KGID, VE	13	0.77	0.53	0.83
Visual Commonsense	ImageNetVC, VCR	6	0.79	0.57	0.59
Average	-	44	0.64	0.55	0.67

C Additional Related Work

Multi-modal Language Models. With the continuous improvement of data scale and computing power, the advancement of Multi-Modal Language Models (MMLMs) has gained momentum.

A OVERVIEW

- 第 B节：零样本多模态评估的详细结果。
- 章节 C：其他相关研究。
- 章节D：传统视觉与语言模型微调的详细结果。
- 章节 E：其他实验及讨论内容。
- 章节 F：指令遵循模型的全面对比分析。
- 章节 G：LLaMA-Adapter与LLAMA-I模型的对比。

B 更多关于多模态评估的细节

ScienceQA（Luet等人。，2022年）评估。如图8所示，ScienceQA数据集中的每个样本都包含视觉上下文、文本上下文、问题、多个选项以及正确答案。为简化分析，我们省略了部分数据样本中的讲解内容。

Zero-shot Multi-modal Evaluation. We test our approach on the three benchmarks (Fu et al., 2023; Liu et al., 2023c; Xu et al., 2023) following their official procedures. In Tables 9, 10 and 11, we respectively report the detailed results of MME and LVLM-eHub benchmarks. As shown, across a wide range of visual question-answering problems, our approach can consistently achieve competitive results. We also show more examples of the multi-modal LLaMA-Adapter for open-domain zero-shot visual questions in Figure 9, where our approach can generate detailed and high-quality responses in natural language. The experiments fully demonstrate the generalization capacity of our proposed multi-modal LLM. We also give some qualitative examples in Figures 9 and 10, where our LLaMA-Adapter can answer open-ended questions for web images.

表9：**在MME基准测试上的感知结果（Fu等人，2023年）**。

模型	ALL	存在性	计数	位置	颜色	海报	名人	场景	地标	艺术作品	OCR
LLaVA	503	50	50	50	55	50	49	50	50	49	50
Minigpt-4模型	867	115	123	82	110	56	65	96	69	56	83
LLaMA-Adapter	973	120	50	48	75	100	86	149	150	70	125

表10：**在MME基准测试上的认知能力测试结果（Fu等人，2023年）**。

模型	ALL	常识推理	数值计算	文本翻译	代码推理
LLaVA	215	57	50	58	50
Minigpt-4	292	72	55	55	110
LLaMA-Adapter	249	81	63	50	55

表11：**LVLM-eHub基准测试下的零样本多模态结果（Xu等人，2023年）**。OC：物体计数；MCI：多类识别；KIE：关键信息提取；VE：视觉蕴含；KGID：基于知识的图像描述；VCR：视觉常识推理任务。

LVLM-eHub基准测试	任务	#数据集	LLaVA	Minigpt-4模型	LLaMA-Adapter
视觉感知	图像分类任务、对象检测、多类别识别	8	0.62	0.73	0.81
视觉知识获取	光学字符识别、知识信息提取、文本生成	17	0.38	0.35	0.44
视觉推理	视觉问答、知识图谱推理任务、视觉推理任务	13	0.77	0.53	0.83
视觉常识	ImageNet视觉常识数据集与视觉常识推理任务	6	0.79	0.57	0.59
平均值	-	44	0.64	0.55	0.67

补充相关研究

多模态语言模型。随着数据规模与计算能力的不断提升，多模态语言模型（MMLMs）的发展势头愈发强劲。

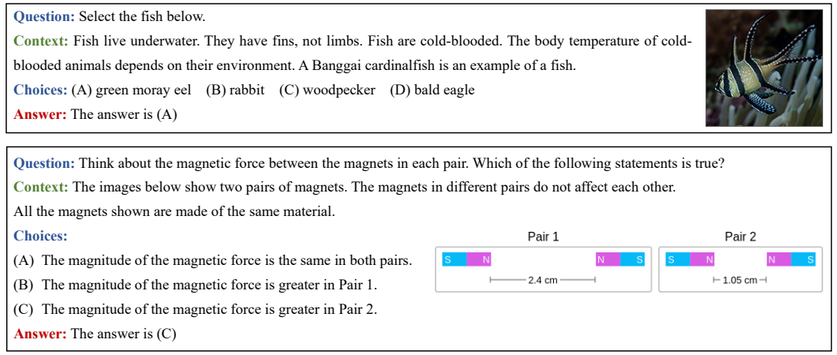


Figure 8: **Multi-modal Reasoning Examples in ScienceQA (Lu et al., 2022).**

Initiatives like CLIP (Radford et al., 2021), ALIGN (Jia et al., 2021), and their derivatives (Li et al., 2022; Gao et al., 2021; Zhai et al., 2022) employ vision-language contrastive pre-training on vast datasets, showcasing robust generalization in zero-shot evaluation. With the rise of LLMs (OpenAI, 2023a;b), modern MMLMs merge these LLM architectures with visual comprehension capacities. BLIP-2 (Li et al., 2023b), for instance, introduces a Q-Former network, bridging frozen image encoders with LLMs. Flamingo (Alayrac et al., 2022) uses interleaved image-text data for few-shot learning, enhancing vision-language inferences. Kosmos (Huang et al., 2023) trains an MMLM on web-scale multi-modal data from scratch, enabling powerful visual perception capacities. While models like Bard (Google, 2023) and GPT-4 (OpenAI, 2023b) remain influential, their closed-source nature has led to the development of MMLMs such as those based on open-source LLaMA (Liu et al., 2023b; Zhu et al., 2023; Ye et al., 2023; Li et al., 2023a; Zhang et al., 2023a). Typically, these MMLMs utilize a two-stage training process. In the initial phase, a substantial quantity of image-text pairs are leveraged to align vision models with LLMs. The subsequent phase involves fine-tuning on a limited set of high-quality datasets to follow human instructions. However, these models are either highly dependent on a fine-tuned instruction model (Vicuna (Chiang et al., 2023) in Mini-GPT4 (Zhu et al., 2023)), or require updating the entire parameters of LLMs (LLaVA (Liu et al., 2023b)). Follow-up works like SPHINX series (Lin et al., 2023; Gao et al., 2024), ImageBind-LLM (Han et al., 2023), Point-LLM (Guo et al., 2023), and MathVerse (Zhang et al., 2024) further explore more diverse potentials of multi-modal capabilities within LLMs. As a concurrent work to LLaVA and Mini-GPT4, our LLaMA-Adapter utilizes zero-initialized attention mechanisms for parameter-efficiently fine-tuning the original LLaMA model, largely saving the expensive full-parameter fine-tuning.

Comparison to Flamingo (Alayrac et al., 2022). As a strong in-context MLLMs, Flamingo adopts a gating strategy for injecting external knowledge into LLMs. Compared to our zero-initialized attention, there are three main differences as follows.

- **Inserted Position.** Our gating works delicately within the self-attention layer of an LLM, more specifically, after the query-key attention scores and before multiplying with value. In contrast, Flamingo's gating is outside and before feeding into LLM's layers, which works right after the newly added cross-attention layer and feed-forward networks.
- **Detailed Mechanism.** Our gating directly reweighs the attention scores of adaption prompts, controlling how much information of prompts is aggregated by the generating word token. Flamingo's gating naively reweighs the residual connection, which controls how much information of visual features is added to all language features.
- **Parameter Efficiency.** Our gating mechanism only introduces 1.2M parameters of efficient learnable prompts. Flamingo's gating is based on newly added large-scale cross-attention layers and FFNs, having over 3B parameters.
- **Application Scenarios.** Due to our lightweight designs, the zero-initialized attention can be adopted either for incorporating language instruction knowledge, or multi-modal image conditions, while Flamingo is specially designed for vision-language tasks by newly adding heavyweight cross-attention modules.

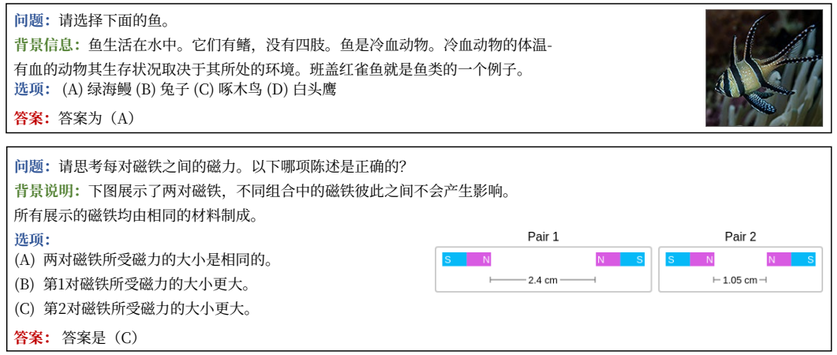


图8：**ScienceQA数据集中的多模态推理示例（卢等人，2022年）。**

Initiatives like CLIP (Radford et al., 2021), ALIGN (Jia et al., 2021), and their derivatives (Li et al., 2022; Gao et al., 2021; Zhai et al., 2022) employ vision-language contrastive pre-training on vast datasets, showcasing robust generalization in zero-shot evaluation. With the rise of LLMs (OpenAI, 2023a;b), modern MMLMs merge these LLM architectures with visual comprehension capacities. BLIP-2 (Li et al., 2023b), for instance, introduces a Q-Former network, bridging frozen image encoders with LLMs. Flamingo (Alayrac et al., 2022) uses interleaved image-text data for few-shot learning, enhancing vision-language inferences. Kosmos (Huang et al., 2023) trains an MMLM on web-scale multi-modal data from scratch, enabling powerful visual perception capacities. While models like Bard (Google, 2023) and GPT-4 (OpenAI, 2023b) remain influential, their closed-source nature has led to the development of MMLMs such as those based on open-source LLaMA (Liu et al., 2023b; Zhu et al., 2023; Ye et al., 2023; Li et al., 2023a; Zhang et al., 2023a). Typically, these MMLMs utilize a two-stage training process. In the initial phase, a substantial quantity of image-text pairs are leveraged to align vision models with LLMs. The subsequent phase involves fine-tuning on a limited set of high-quality datasets to follow human instructions. However, these models are either highly dependent on a fine-tuned instruction model (Vicuna (Chiang et al., 2023) in Mini-GPT4 (Zhu et al., 2023)), or require updating the entire parameters of LLMs (LLaVA (Liu et al., 2023b)). Follow-up works like SPHINX series (Lin et al., 2023; Gao et al., 2024), ImageBind-LLM (Han et al., 2023), Point-LLM (Guo et al., 2023), and MathVerse (Zhang et al., 2024) further explore more diverse potentials of multi-modal capabilities within LLMs. As a concurrent work to LLaVA and Mini-GPT4, our LLaMA-Adapter utilizes zero-initialized attention mechanisms for parameter-efficiently fine-tuning the original LLaMA model, largely saving the expensive full-parameter fine-tuning.

与Flamingo的对比（Alayrac等人。、2022年）。作为优秀的上下文内大语言模型，Flamingo采用了门控策略将外部知识注入到大语言模型中。与我们所采用的零初始化注意力机制相比，二者存在以下三个主要差异。

- **插入位置**。我们的门控机制精细地设置在大语言模型的自注意力层内部，更确切地说，是在查询键注意力分数计算之后、与值向量相乘之前。相比之下，Flamingo模型的门控机制则位于模型层次结构之外、输入自注意力层之前，即在新添加的交叉注意力层和前馈网络之后立即发挥作用。
- **详细机制**。我们的门控机制会直接重新调整适配提示词的注意力分数，从而控制生成词标记会聚合多少来自这些提示词的信息。而Flamingo模型的门控机制则是直接重新调整残差连接，以此决定将多少视觉特征信息加入所有的语言特征中。
- **参数效率**。我们的门控机制仅引入了120万参数的高效可学习提示语。而Flamingo模型的门控机制则是基于新添加的大规模交叉注意力层与FFN结构，这类结构的参数量超过30亿。
- **应用场景**。得益于我们轻量级的设计，零初始化注意力机制既可用于整合语言指令知识，也可用于处理多模态图像信息；而Flamingo则通过新增重量较大的跨注意力模块，专为视觉语言任务而设计。

Can you introduce me this movie?



Titanic is a 1997 epic romantic disaster film directed by James Cameron. The film is based on the real-life events of the sinking of the RMS Titanic, a British passenger liner that sank in the North Atlantic Ocean in 1912 after colliding with an iceberg during its maiden voyage from Southampton to New York City. The movie stars Leonardo DiCaprio and Kate Winslet as the two main characters, Jack Dawson and Rose DeWitt Bukater, respectively. The film follows the story of their love affair and the events leading up to the sinking of the Titanic. The movie is known for its stunning visual effects, captivating storyline, and memorable performances by its lead actors. Titanic received widespread critical acclaim and was a commercial success, becoming the highest-grossing film of all time until it was surpassed by Cameron's own Avatar in 2010. The film's legacy continues to endure, with its iconic scenes, memorable quotes, and timeless love story inspiring generations of moviegoers.

Why this image looks funny?



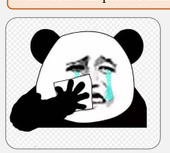
The image looks funny because it is an unconventional and unexpected scene. It features a group of animals, including a cat, a dog, a chicken and a donkey, all sitting on top of each other. This is not a typical interaction between these animals, as they usually do not interact with each other in such a manner. The fact that they are all sitting on top of one another creates a humorous and amusing visual, as it is not something one would typically see in a natural setting.

What message does this picture want to convey?



The picture wants to convey a message of unity, togetherness, and connection among the people in the group. The image shows a large group of people standing in a circle, with their feet touching each other. This arrangement creates a sense of closeness and shared experience among the individuals, emphasizing the importance of community and collaboration. The act of touching feet can symbolize trust, support, and mutual understanding, which are essential elements of a strong and cohesive group.

What does this picture mean in online chat?



In online chat, the picture of a panda crying with its mouth open can be interpreted as a sad or emotional expression. It can be used to convey feelings of sadness, disappointment, or even sympathy. The panda's facial expression and the use of a panda, which is often associated with cuteness, can evoke a sense of empathy and understanding from the recipient.

Figure 9: **Zero-shot Multi-modal Understanding Examples of LLaMA-Adapter: Part 1.**

能给我介绍一下这部电影吗?



《泰坦尼克号》是1997年由詹姆斯·卡梅隆执导的一部史诗级浪漫灾难电影。该片以泰坦尼克号实际沉没事件为蓝本创作。
RMS泰坦尼克号是一艘英国客轮，1912年在北大西洋航行时与冰山相撞后沉没。
从南安普顿出发前往纽约市的处女航。这部电影由莱昂纳多·迪卡普里奥与凯特·温斯莱特领衔主演，饰演这两位主角。
角色分别对应杰克·道森与罗丝·德威特·布卡特。这部电影讲述了他们之间的爱情故事以及随后发生的一系列事件。
在泰坦尼克号沉没之前。这部电影以其令人惊叹的视觉效果、扣人心弦的故事情节以及难以忘怀的
其主演们的出色表现也为影片增色不少。《泰坦尼克号》不仅获得了业界的高度赞誉，更在商业上大获成功，一度成为有史以来票房最高的电影。
这一纪录一直保持到2010年，才被卡梅隆自己打造的《阿凡达》超越。尽管如此，这部电影的传奇影响力依然持续存在。
影片中那些经典的场景、令人难忘的台词以及跨越时光的爱情故事，一直激励着一代又一代的电影爱好者。

为什么这张图片看起来很有趣?



这张图片之所以显得有趣，是因为它呈现了一个非常规且出人意料的场景：有一群动物，包括一只猫、一只狗、一只鸡和一头驴，全都挤在一起。
这种动物之间的互动方式并不常见，因为它们平时通常不会以这种方式相处。
它们不会以这种方式相互接触。由于全都挤作一团，便形成了一种既幽默又
有趣的画面，因为这种场景在自然环境中是极为罕见的。

这张图片想要传达怎样的信息呢?



这张图片旨在表达群体中人们之间的团结、和睦与紧密联系。画面展示的是
一大群人围成一圈站立，彼此的脚相互触碰。这样的排列营造出一种亲密感，
同时也凸显出个体之间共享经历的重要性，强调了社群与协作的重要意义。脚部相触这一动作
这些元素能够象征信任、支持与相互理解，而这些都是构建一个坚强且团结的群体的重要要素。

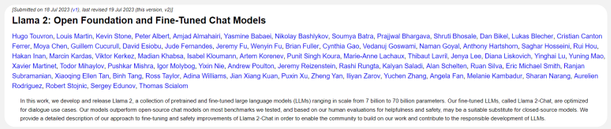
在网络聊天中，这张图片到底代表着什么含义呢?



在网络聊天里，那只张着嘴哭泣的熊猫图片可以被理解为一种悲伤或情绪化的表达，人们可以借此来
传达悲伤、失望甚至同情的情感。熊猫的面部表情，再加上通常被认为充满萌态的熊猫形象，
能够让接收信息的人产生共鸣与理解。

图9：**LLaMA-Adapter的零样本多模态理解示例——第一部分。**

What is Llama2?



[Submitted on 18 Jul 2023 (v1), last revised 19 Jul 2023 (this version, v2)]

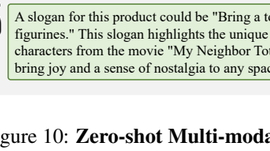
Llama 2: Open Foundation and Fine-Tuned Chat Models

Hugo Touvron, Louis Martin, Kevin Stone, Peter Albert, Amjad Almahairi, Yasmine Babaei, Nikolay Bashlykov, Soumya Batra, Prajjwal Bhargava, Shruti Bhosale, Dan Bikel, Lukas Blecher, Cristian Canton Ferrer, Moya Chen, Guillem Cucurull, David Esiobu, Jude Fernandes, Jeremy Fu, Wenyin Fu, Brian Fuller, Cynthia Gao, Vedanuj Goswami, Naman Goyal, Anthony Hartshorn, Saghar Hosseini, Rui Hou, Hakan Inan, Marcin Kardas, Viktor Kerkez, Madian Khabsa, Isabel Kloumann, Artem Korenev, Punit Singh Koura, Marie-Anne Lachaux, Thibaut Lavril, Jenya Lee, Diana Liskovich, Yinghai Lu, Yuning Mao, Xavier Martinet, Todor Mihaylov, Pushkar Mishra, Igor Molybog, Yixin Nie, Andrew Poulton, Jeremy Reizenstein, Rashi Rungta, Kalyan Saladi, Alan Schelten, Ruan Silva, Eric Michael Smith, Ranjan Subramanian, Xiaoqing Ellen Tan, Binh Tang, Ross Taylor, Adina Williams, Jian Xiang Kuan, Puxin Xu, Zheng Yan, Iliyan Zarov, Yuchen Zhang, Angela Fan, Melanie Kambadur, Sharan Narang, Aurelien Rodriguez, Robert Stojnic, Sergey Edunov, Thomas Scialom

In this work, we develop and release Llama 2, a collection of pretrained and fine-tuned large language models (LLMs) ranging in scale from 7 billion to 70 billion parameters. Our fine-tuned LLMs, called Llama 2-Chat, are optimized for dialogue use cases. Our models outperform open-source chat models on most benchmarks we tested, and based on our human evaluations for helpfulness and safety, may be a suitable substitute for closed-source models. We provide a detailed description of our approach to fine-tuning and safety improvements of Llama 2-Chat in order to enable the community to build on our work and contribute to the responsible development of LLMs.

Llama2 is a software model that is used for language learning and machine learning research. It is an open-source implementation of the Llama model, which is a neural network-based language model that can be used for tasks such as text generation, text classification, and language translation.

What should I do when my computer shows up with this screen?



When your computer displays a blue screen, it typically indicates a software or hardware issue that needs to be addressed. Here are some steps you can take to resolve the issue:
1. Restart your computer: Restarting your computer will clear any temporary files or issues that may be causing the blue screen error.
2. Check for updates: Ensure that your computer is up to date with the latest software and drivers. This can help resolve issues related to outdated software or hardware compatibility.
3. Run a system scan: Use your computer's built-in diagnostic tools or a third-party antivirus and malware scanner to check for any issues that may be causing the blue screen error.
4. Contact technical support: If the issue persists after following these steps, you may need to contact your computer manufacturer's technical support for further assistance.
5. Backup your data: Make sure to back up your important files and data to an external storage device or cloud storage to avoid losing any important information in case the issue cannot be resolved.

What is wrong with my bike? How can I fix it?



In the image, there is a bicycle with a flat tire. To fix this issue, you should first check if the tire is punctured or if there is a problem with the valve. If the tire is punctured, you can use a bicycle pump to inflate it, and then patch the hole with a tire repair kit or a piece of inner tube. If the valve is the issue, you can replace it with a new one or use a bicycle valve tool to remove the old one and insert a new one. Once the tire is repaired or replaced, you can reattach the wheel to the bicycle frame and continue riding.

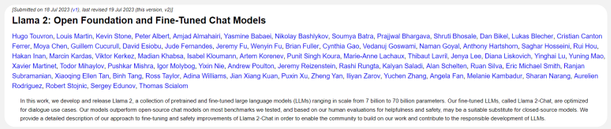
Design a slogan for this product



A slogan for this product could be "Bring a touch of whimsy and charm to your space with our adorable Totoro and rabbit figurines." This slogan highlights the unique and charming design of the figurines, which are inspired by the popular animated characters from the movie "My Neighbor Totoro." The slogan emphasizes the appeal of these figurines as decorative items that can bring joy and a sense of nostalgia to any space, whether it's a home, office, or store.

Figure 10: **Zero-shot Multi-modal Understanding Examples of LLaMA-Adapter: Part 2.**

什么是Llama2?



[Submitted on 18 Jul 2023 (v1), last revised 19 Jul 2023 (this version, v2)]

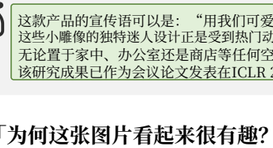
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In this work, we develop and release Llama 2, a collection of pretrained and fine-tuned large language models (LLMs) ranging in scale from 7 billion to 70 billion parameters. Our fine-tuned LLMs, called Llama 2-Chat, are optimized for dialogue use cases. Our models outperform open-source chat models on most benchmarks we tested, and based on our human evaluations for helpfulness and safety, may be a suitable substitute for closed-source models. We provide a detailed description of our approach to fine-tuning and safety improvements of Llama 2-Chat in order to enable the community to build on our work and contribute to the responsible development of LLMs.

Llama2是一种用于语言学习及机器学习研究的软件模型，它属于基于开源实现的版本。
它是基于神经网络构建的LLaMA模型的开源实现，可应用于文本生成、文本分类以及语言翻译等各类任务。
此外它还能用于其他相关的语言处理任务。

当我的电脑出现这样的屏幕时，我该怎么做?



当电脑显示蓝屏时，通常意味着存在需要解决的软件或硬件问题，以下是一些解决该问题的方法。
具体可以采取以下步骤来解决问题：
1. 重启您的电脑：重启电脑可以清除可能导致蓝屏的任何临时文件或故障。
错误。
2. 检查更新：请确保您的电脑已安装最新的软件和驱动程序，这有助于解决各种问题。
与过时的软件或硬件兼容性有关。
3. 运行系统扫描：使用电脑内置的诊断工具或第三方杀毒软件及恶意程序扫描工具，检查是否存在任何问题
可能导致蓝屏错误的各种问题。
4. 联系技术支持：如果按照这些步骤操作后问题依然存在，您可能需要联系您的电脑厂商。
制造商的技术支持团队可提供进一步协助。
5. 备份数据：务必将重要文件和数据备份到外部存储设备或云存储中，以避免丢失。
以免在问题无法解决时丢失任何重要信息。

我的自行车怎么了？我该怎么修理它?



在图片中，有一辆轮胎漏气的自行车。要解决这个问题，你首先应该检查轮胎是否出现了穿孔或其他故障。
连同气门芯一起。如果轮胎出现穿孔，可以使用自行车打气筒给它充气，然后用轮胎修补套件将孔洞补上。
一段内胎。如果问题出在气门芯上，您可以更换一个新的，或者使用自行车气门芯拆卸工具把旧的气门芯取出来。
插入一个新的。一旦轮胎得到修复或更换，就可以再将车轮装回自行车车架，继续骑行了。

为这款产品设计一句宣传语



这款产品的宣传语可以是：“用我们可爱的龙猫与兔子，为您的空间增添一丝奇思妙想与魅力”
这些小雕像的独特迷人设计正是受到热门动画电影《龙猫》中角色的启发。这句宣传语凸显了它们作为装饰品的吸引力。
无论置于家中、办公室还是商店等任何空间，这些小雕像都能带来欢乐与怀旧之情。
该研究成果已作为会议论文发表在ICLR 2024会议上。

「为何这张图片看起来很有趣?」

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D MORE DETAILED RESULTS OF MODEL FINE-TUNING

In this section, we provide more detailed experiments and analysis of applying our zero-initialized attention to fine-tune vision models, language models, and vision-language models, respectively.

D.1 DETAILED RESULTS ON VISION MODELS

In Table 12, we compare the detailed fine-tuning results on VTAB-1k (Zhai et al., 2019) benchmark with 19 downstream visual tasks, which can be categorized into Natural (7 tasks), Specialized (4 tasks), and Structured (8 tasks), according to image domains. As shown, our zero-initialized attention outperforms VPT (Jia et al., 2022) on most datasets (16 out of 19), and surpasses full fine-tuning along with other fine-tuning methods by large margins. This demonstrates the general efficacy of the proposed mechanism on a variety of image distributions.

Table 12: **Detailed Fine-tuning Results on VTAB-1k Benchmark.** We report the top-1 accuracy and adopt ViT-B/16 (Dosovitskiy et al., 2020) pre-trained on supervised ImageNet-21k (Deng et al., 2009) as the base model. We compare our zero-initialized attention with Bias (Zaken et al., 2022), Adapter (Houlsby et al., 2019), Sidetune (Zhang et al., 2020) and VPT (Jia et al., 2022).

	CIFAR100	Caltech101	DTD	Flowers102	OxfordPets	SVHN	SUN397	**Mean**	Patch Camelyon	EuroSAT	Resisc45	Retinopathy	**Mean**	Clevr/count	Clevr/distance	DMLab	KITTI/distance	dSprites/location	dSprites/orientation	SmallNORB/azimuth	SmallNORB/elevation	**Mean**
Full	68.9	87.7	64.3	97.2	86.9	87.4	38.8	75.9	79.7	95.7	84.2	73.9	83.4	56.3	58.6	41.7	65.5	57.5	46.7	25.7	29.1	47.6
Bias	72.8	87.0	59.2	97.5	85.3	59.9	51.4	73.3	78.7	91.6	72.9	69.8	78.3	61.5	55.6	32.4	55.9	66.6	40.0	15.7	25.1	44.1
Adapter	74.1	85.7	62.7	97.8	87.2	34.6	50.7	70.4	76.3	87.5	73.7	70.9	77.1	45.2	41.8	31.2	56.4	31.9	25.4	13.5	22.0	33.4
Sidetune	60.7	60.8	53.6	95.5	66.7	34.9	35.3	58.2	58.5	87.7	65.2	61.0	68.1	27.6	22.6	31.3	51.7	8.2	14.4	9.8	21.8	23.4
VPT	78.8	90.8	65.8	98.0	88.3	78.1	49.6	78.5	81.8	96.1	83.4	68.4	82.4	68.5	60.0	46.5	72.8	73.6	47.9	32.9	37.7	55.0
Zero-init.	82.2	92.4	70.3	98.4	89.8	84.9	54.3	**81.7**	83.6	95.3	85.0	73.8	**84.4**	69.3	60.2	51.1	79.7	80.7	49.0	30.6	33.6	**56.8**

D.2 MORE EXPERIMENTS ON LANGUAGE TASKS

For a more comprehensive evaluation of zero-initialized attention, we fine-tune RoBERTa$_{\text{large}}$ (Liu et al., 2019) on other two natural language processing tasks in addition to extractive question answering of the main paper, which are named entity recognition (NER) and semantic role labeling (SRL). We adopt CoNLL03 (Sang & De Meulder, 2003), CoNLL04 (Carreras & Màrquez, 2004), CoNLL05 (Carreras & Màrquez, 2005), and CoNLL12 (Pradhan et al., 2012) as the evaluation datasets. As shown in Table 13, compared to P-tuning V2 (PT2) (Liu et al., 2021a), our zero-initialized attention can steadily perform better on all datasets with varying magnitudes, which indicates our effectiveness for different language tasks and applications.

Table 13: **Language Model Fine-tuning** with RoBERTa$_{\text{large}}$ (Liu et al., 2019) on named entity recognition (NER) and semantic role labeling (SRL) tasks. We report the micro-f1 score. * denotes our reproduced results.

Method	CoNLL03	CoNLL04	CoNLL12	CoNLL05$_{Brown}$	CoNLL05$_{WSJ}$
Full	92.6	88.8	86.5	85.6	90.2
PT (Lester et al., 2021)	86.1	76.2	67.2	70.7	76.8
PT2 (Liu et al., 2021a)	92.8	88.4	84.6	84.3	89.2
PT2*	91.8	88.4	84.7	83.9	89.4
Zero-init.	**92.4**	**88.8**	**85.2**	**84.7**	**89.6**

D.3 DETAILED RESULTS ON VISION-LANGUAGE MODELS

Besides ViT and RoBERTa, we also evaluate our approach on CLIP (Radford et al., 2021), a vision-language model pre-trained by 400 million text-image pairs. In detail, we adopt CLIP with a ViT-B/16 as the visual encoder and a 12-layer transformer (Li et al., 2019b) as the textual encoder. We test our

D 模型微调的更详细结果

在本节中，我们将针对将所设计的零初始化注意力机制分别应用于视觉模型、语言模型以及视觉语言模型的微调过程，展开更为详尽的实验与分析。

D.1 视觉模式下的详细结果

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In Table 12, we compare the detailed fine-tuning results on VTAB-1k (Zhai et al., 2019) benchmark with 19 downstream visual tasks, which can be categorized into Natural (7 tasks), Specialized (4 tasks), and Structured (8 tasks), according to image domains. As shown, our zero-initialized attention outperforms VPT (Jia et al., 2022) on most datasets (16 out of 19), and surpasses full fine-tuning along with other fine-tuning methods by large margins. This demonstrates the general efficacy of the proposed mechanism on a variety of image distributions.

Table 12: **Detailed Fine-tuning Results on VTAB-1k Benchmark.** We report the top-1 accuracy and adopt ViT-B/16 (Dosovitskiy et al., 2020) pre-trained on supervised ImageNet-21k (Deng et al., 2009) as the base model. We compare our zero-initialized attention with Bias (Zaken et al., 2022), Adapter (Houlsby et al., 2019), Sidetune (Zhang et al., 2020) and VPT (Jia et al., 2022).

	CIFAR100	Caltech101	DTD	Flowers102	OxfordPets	SVHN	SUN397	**Mean**	Patch Camelyon	EuroSAT	Resisc45	Retinopathy	**Mean**	Clevr/count	Clevr/distance	DMLab	KITTI/distance	dSprites/location	dSprites/orientation	SmallNORB/azimuth	SmallNORB/elevation	**Mean**
Full	68.9	87.7	64.3	97.2	86.9	87.4	38.8	75.9	79.7	95.7	84.2	73.9	83.4	56.3	58.6	41.7	65.5	57.5	46.7	25.7	29.1	47.6
Bias	72.8	87.0	59.2	97.5	85.3	59.9	51.4	73.3	78.7	91.6	72.9	69.8	78.3	61.5	55.6	32.4	55.9	66.6	40.0	15.7	25.1	44.1
适配器	74.1	85.7	62.7	97.8	87.2	34.6	50.7	70.4	76.3	87.5	73.7	70.9	77.1	45.2	41.8	31.2	56.4	31.9	25.4	13.5	22.0	33.4
侧向微调	60.7	60.8	53.6	95.5	66.7	34.9	35.3	58.2	58.5	87.7	65.2	61.0	68.1	27.6	22.6	31.3	51.7	8.2	14.4	9.8	21.8	23.4
VPT	78.8	90.8	65.8	98.0	88.3	78.1	49.6	78.5	81.8	96.1	83.4	68.4	82.4	68.5	60.0	46.5	72.8	73.6	47.9	32.9	37.7	55.0
零初始化。	82.2	92.4	70.3	98.4	89.8	84.9	54.3	**81.7**	83.6	95.3	85.0	73.8	**84.4**	69.3	60.2	51.1	79.7	80.7	49.0	30.6	33.6	**56.8**

D.2 更多语言任务相关的实验

For a more comprehensive evaluation of zero-initialized attention, we fine-tune RoBERTa$_{\text{large}}$ (Liu et al., 2019) on other two natural language processing tasks in addition to extractive question answering of the main paper, which are named entity recognition (NER) and semantic role labeling (SRL). We adopt CoNLL03 (Sang & De Meulder, 2003), CoNLL04 (Carreras & Màrquez, 2004), CoNLL05 (Carreras & Màrquez, 2005), and CoNLL12 (Pradhan et al., 2012) as the evaluation datasets. As shown in Table 13, compared to P-tuning V2 (PT2) (Liu et al., 2021a), our zero-initialized attention can steadily perform better on all datasets with varying magnitudes, which indicates our effectiveness for different language tasks and applications.

表13：**使用RoBERTa模型的语言模型微调**技术$_{\text{large}}$ （Liu等人，2019年）在命名实体识别与语义角色标注任务上的应用。我们报告了对应的F1分数，其中带*标记的结果为我们在相同条件下重新得到的实验结果。

方法	CoNLL03	CoNLL04	CoNLL12	CoNLL05$_{Brown}$	CoNLL05$_{WSJ}$
Full	92.6	88.8	86.5	85.6	90.2
PT (Lester等人，2021年)	86.1	76.2	67.2	70.7	76.8
PT2 (Liu等人，2021a)	92.8	88.4	84.6	84.3	89.2
PT2*	91.8	88.4	84.7	83.9	89.4
零初始化。	**92.4**	**88.8**	**85.2**	**84.7**	**89.6**

D.3 视觉语言模型的详细结果

除了ViT和RoBERTa之外，我们还在CLIP（Radford等人，2021年）上对我们的方法进行了评估，CLIP是一种通过4亿对文本-图像对预训练的视觉语言模型。具体而言，我们采用的CLIP配置为以ViT-B/16作为视觉编码器，以12层变换器模型（李等人，2019b）作为文本编码器。我们对此进行了测试

Table 14: **Vision-Language Model Fine-tuning** with ViT-B/16 CLIP (Radford et al., 2021) on base-to-novel generalization (Zhou et al., 2022b) benchmark. We report the classification accuracy (%) and harmonic mean (HM).

Method	ImageNet			Caltech101			Flowers102			Average		
	Base	Novel	HM	Base	Novel	HM	Base	Novel	HM	Base	Novel	HM
CLIP (Radford et al., 2021)	72.43	68.14	70.22	96.84	94.00	95.40	72.08	77.80	74.83	80.45	79.98	80.15
CoOp (Zhou et al., 2022c)	76.47	67.88	71.92	98.00	89.81	93.73	97.60	59.67	74.06	90.69	72.45	79.90
CoCoOp (Zhou et al., 2022b)	75.98	70.43	73.10	97.96	93.81	95.84	94.87	71.75	81.71	89.60	78.66	83.55
MaPLe (Khattak et al., 2022)	76.66	70.54	73.47	97.74	94.36	96.02	95.92	72.46	82.56	90.11	79.12	84.02
Zero-init.	**76.70**	**71.00**	**73.74**	**98.10**	**94.53**	**96.28**	**96.00**	**74.67**	**84.00**	**90.27**	**80.07**	**84.67**

fine-tuning results on base-to-novel generalization (Zhou et al., 2022b) benchmark with three datasets, i.e., ImageNet (Deng et al., 2009), Caltech101 (Fei-Fei et al., 2004), and Flowers102 (Nilsback & Zisserman, 2008), where the model is trained only on the base classes in a few-shot setting and evaluated on both base and novel categories. We freeze the entire CLIP and insert the adaption prompts with zero-initialized attention into CLIP's encoders. As shown in Table 14, our approach achieves the best average classification accuracy on both base and novel categories, demonstrating our fine-tuning capability for large vision-language models.

E Additional Experiment and Discussion

E.1 Evaluation on Counterfactual Reasoning

As a core ability of human intelligence, counterfactual reasoning is a challenging assessment for multi-modal LLMs, which involves the processing of alternatives to observed states or past events. Here, we adopt the very recent C-VQA (Zhang et al., 2023b) benchmark for evaluating our counterfactual reasoning capability. C-VQA contains 2K counterfactual question and answer pairs, which are collected from VQAv2 (Goyal et al., 2017) and supplemented by ChatGPT (OpenAI, 2023a). As shown in Table 15, for three groups of questions, LLaMA-Adapter performs comparably to the concurrent LLaVA. Especially for the 'Numerical indirect' questions, our approach achieves the best counterfactual reasoning results (34.3) and the least performance loss (5.6↓) than all other models.

Table 15: **Counterfactual Reasoning Evaluation on C-VQA (Zhang et al., 2023b) Benchmark.**

Method	Numerical direct↑ (Loss↓)	Numerical indirect↑ (Loss↓)	Boolean↑ (Loss↓)
ViperGPT (Surís et al., 2023)	80.6 (19.4↓)	31.6 (68.4↓)	21.6 (72.4↓)
LLaVA-7B (Liu et al., 2023b)	27.0 (9.9↓)	25.0 (15.2↓)	58.5 (4.8↓)
LLaVA-13B (Liu et al., 2023b)	24.8 (11.9↓)	20.8 (21.2↓)	56.3 (4.7↓)
LLaMA-Adapter-7B	30.1 (5.8↓)	34.3 (5.6↓)	45.8 (14.5↓)

E.2 Evaluation on Object Hallucination

Similar to language generation, multi-modal LLMs also suffer from the hallucination issue, i.e., they might generate descriptions containing objects inconsistent with the target images. To validate our approach's performance, we adopt POPE (Li et al., 2023c) for object hallucination evaluation, which converts the object hallucination problem as a binary classification task and includes 500 images from MSCOCO (Lin et al., 2014) with 6 questions per sample. As shown in Table 16, for different evaluation settings, LLaMA-Adapter with LLaMA-7B attains competitive accuracy compared to other multi-modal LLMs with LLaMA-13B, which indicates our relatively stronger robustness to object hallucination problems.

E.3 Tuning by More Instruction Data

By default, we utilize a combination of Alpaca's data (52K) (Taori et al., 2023) and LLaVA-I (158K) Liu et al. (2023b) for visual instruction tuning. Here, we progressively add more

表14：**基于ViT-B/16 CLIP模型的视觉语言模型微调**（Radford等人，2021年）在基础到新颖样本泛化（Zhou等人，2022b）基准测试上的表现。文中报告了分类准确率（%）与调和平均数（HM）。

方法	ImageNet			Caltech101			Flowers102			平均值		
	Base	新型	HM	Base	新型	HM	Base	新型	HM	Base	新型	HM
CLIP模型（Radford等人，2021年）	72.43	68.14	70.22	96.84	94.00	95.40	72.08	77.80	74.83	80.45	79.98	80.15
CoOp（Zhou等人，2022c）	76.47	67.88	71.92	98.00	89.81	93.73	97.60	59.67	74.06	90.69	72.45	79.90
CoCoOp（Zhou等人，2022b）	75.98	70.43	73.10	97.96	93.81	95.84	94.87	71.75	81.71	89.60	78.66	83.55
MaPLe（Khattak等人，2022年）	76.66	70.54	73.47	97.74	94.36	96.02	95.92	72.46	82.56	90.11	79.12	84.02
零初始化。	**76.70**	**71.00**	**73.74**	**98.10**	**94.53**	**96.28**	**96.00**	**74.67**	**84.00**	**90.27**	**80.07**	**84.67**

fine-tuning results on base-to-novel generalization (Zhou et al., 2022b) benchmark with three datasets, i.e., ImageNet (Deng et al., 2009), Caltech101 (Fei-Fei et al., 2004), and Flowers102 (Nilsback & Zisserman, 2008), where the model is trained only on the base classes in a few-shot setting and evaluated on both base and novel categories. We freeze the entire CLIP and insert the adaption prompts with zero-initialized attention into CLIP's encoders. As shown in Table 14, our approach achieves the best average classification accuracy on both base and novel categories, demonstrating our fine-tuning capability for large vision-language models.

E 其他实验与讨论

E.1 对反事实推理能力的评估

As a core ability of human intelligence, counterfactual reasoning is a challenging assessment for multi-modal LLMs, which involves the processing of alternatives to observed states or past events. Here, we adopt the very recent C-VQA (Zhang et al., 2023b) benchmark for evaluating our counterfactual reasoning capability. C-VQA contains 2K counterfactual question and answer pairs, which are collected from VQAv2 (Goyal et al., 2017) and supplemented by ChatGPT (OpenAI, 2023a). As shown in Table 15, for three groups of questions, LLaMA-Adapter performs comparably to the concurrent LLaVA. Especially for the 'Numerical indirect' questions, our approach achieves the best counterfactual reasoning results (34.3) and the least performance loss (5.6↓) than all other models.

表15：**基于C-VQA的假设推理评估（Zhang等人，2023b年）基准测试。**

方法	数值型直接↑ (损失↓)	数值型间接↑ (损失↓)	布尔型↑ (损失↓)
ViperGPT (Surís 等人。，2023年)	80.6 (19.4↓)	31.6（68.4↓）	21.6（72.4↓）
LLaVA-70亿参数（Liu等人，2023b年）	27.0（9.9↓）	25.0（15.2↓）	58.5（4.8↓）
LLaVA-13B（Liu等人，2023b年）	24.8（11.9↓）	20.8（21.2↓）	56.3（4.7↓）
LLaMA-Adapter-7B	30.1（5.8↓）	34.3（5.6↓）	45.8（14.5↓）

E.2 对对象幻觉的评估

Similar to language generation, multi-modal LLMs also suffer from the hallucination issue, i.e., they might generate descriptions containing objects inconsistent with the target images. To validate our approach's performance, we adopt POPE (Li et al., 2023c) for object hallucination evaluation, which converts the object hallucination problem as a binary classification task and includes 500 images from MSCOCO (Lin et al., 2014) with 6 questions per sample. As shown in Table 16, for different evaluation settings, LLaMA-Adapter with LLaMA-7B attains competitive accuracy compared to other multi-modal LLMs with LLaMA-13B, which indicates our relatively stronger robustness to object hallucination problems.

E.3 通过更多指令数据进行调优

默认情况下，我们在视觉指令微调中采用了Alpaca模型对应的数据（52K）（Taori等人，2023年）与LLaVA-I模型对应的数据（158K）Liu等人（2023b年）相结合的方式。在此过程中，我们会逐步增加更多

Table 16: **Object Hallucination Evaluation on POPE (Li et al., 2023d) Benchmark.**

Method	Random	Popular	Adversarial
InstructBLIP-13B (Dai et al., 2023b)	88.73	81.37	74.37
mPLUG-Owl-7B (Ye et al., 2023)	53.30	50.63	50.67
LLaVA-13B (Liu et al., 2023b)	54.43	52.43	50.77
MM-GPT-7B (Gong et al., 2023)	50.03	50.00	50.00
LLaMA-Adapter-7B	75.47	60.43	60.66

question-answering data to enlarge the instruction datasets of LLaMA-Adapter: the sampled 83K VQAv2 (Goyal et al., 2017) by LLaVA-1.5 (Liu et al., 2023a) and the entire 204K VQAv2. We also compare our performance with very recent multi-modal LLMs with advanced visual reasoning capabilities: InstructBLIP (Dai et al., 2023a) and LLaVA-1.5. InstructBLIP collects extensive visual question-answering datasets (16M) to fine-tune BLIP-2 (Li et al., 2023b), which endows robust visual instruction-following capabilities. LLaVA-1.5 is an upgraded variant of LLaVA with a more powerful LLM, i.e., LLaMA-2 (Touvron et al., 2023), and is also fine-tuned by a collection of 665K instruction-tuning datasets. As shown in Table 17, the increasing instruction tuning data leads to better multi-modal reasoning results on three benchmarks, demonstrating our method's favorable scalability to data size. Our LLaMA-Adapter also achieves comparable performance to the latest InstructBLIP and LLaVA-1.5, further indicating our effectiveness for multi-modal reasoning.

Table 17: **Instruction-tuning with More Datasets** on three zero-shot multi-modal Benchmarks: MME (Fu et al., 2023), MMBench (Liu et al., 2023c), and LVLM-eHub (Xu et al., 2023).

Model	MME			MMBench							LVLM-eHub				
	All	P	C	All	LR	AR	RR	FP-S	FP-C	CP	All	VP	VKA	VR	VC
BLIP-2	1584	1294	290	-	-	-	-	-	-	-	0.77	0.86	0.93	0.76	0.54
InstructBLIP	1505	1213	292	33.9	21.6	47.4	22.5	33.0	24.4	41.1	0.95	0.93	0.97	0.91	0.99
MiniGPT-4	1159	867	292	23.0	13.6	32.9	8.9	28.7	11.2	28.3	0.55	0.73	0.35	0.53	0.57
LLaVA	718	503	215	36.2	15.9	53.6	28.6	41.8	20.0	40.4	0.64	0.62	0.38	0.77	0.79
LLaVA-1.5	1826	1531	295	59.5	32.4	72.6	49.3	62.3	52.2	67.7	-	-	-	-	-
LLaMA-Adapter	1222	973	249	39.5	13.1	47.4	23.0	45.0	33.2	50.6	0.6675	0.81	0.44	0.83	0.59
+VQAv2 (83K)	1256	1007	249	43.4	22.9	44.7	31.3	46.7	46.9	50.3	0.6925	0.84	0.42	0.88	0.63
+VQAv2 (204K)	1618	1272	346	60.1	34.7	65.3	48.7	63.1	57.3	69.3	0.7175	0.86	0.44	0.92	0.65

E.4 More Quantitative Comparison with Alpaca-LoRA

Besides qualitative results, We have compared the language generative capabilities of our LLaMA-Adapter, Alpaca (Taori et al., 2023), and Alpaca-LoRA (alp, 2023) on the GPT-4 evaluation benchmark (Chiang et al., 2023) in Figure 5, which utilizes GPT-4 to assess the response quality on 80 questions. Here, we further evaluate the language processing capacity of the three methods on Open LLM benchmark (Beeching et al., 2023). It evaluates LLMs' generative abilities in four different tasks: AI2 Reasoning Challenge (Clark et al., 2018), HellaSwag (Zellers et al., 2019), MMLU (Hendrycks et al., 2021), and TruthfulQA (Lin et al., 2022). Each task contains challenging data samples over a wide range of knowledge domains. As shown in Table 18, LLaMA-Adapter still achieves the best average performance than Alpaca's full fine-tuning and Alpaca-LoRA. This demonstrates the strong language instruction-following ability of our approach.

Table 18: **Quantitative Evaluation on Open LLM Benchmark (Beeching et al., 2023).**

Method	Avg	ARC	HellaSwag	MMLU	TruthfulQA
Alpaca (Taori et al., 2023)	49.23	49.1	77.7	33.8	36.3
Alpaca-LoRA (alp, 2023)	50.73	53	77.9	37.1	34.9
LLaMA-Adapter	52.2	54.7	78.8	34.9	40.4

表16：**基于POPE算法的物体幻觉评估（李等人，2023年）基准测试。**

方法	随机	流行	对抗性
InstructBLIP-13B（Dai等人。，2023b年）	88.73	81.37	74.37
mPLUG-Owl-70亿参数（Ye等人。，2023年）	53.30	50.63	50.67
LLaVA-13B（Liu等人，2023b年）	54.43	52.43	50.77
MM-GPT-7B（Gong等人。、2023年）	50.03	50.00	50.00
LLaMA-Adapter-70亿参数	75.47	60.43	60.66

question-answering data to enlarge the instruction datasets of LLaMA-Adapter: the sampled 83K VQAv2 (Goyal et al., 2017) by LLaVA-1.5 (Liu et al., 2023a) and the entire 204K VQAv2. We also compare our performance with very recent multi-modal LLMs with advanced visual reasoning capabilities: InstructBLIP (Dai et al., 2023a) and LLaVA-1.5. InstructBLIP collects extensive visual question-answering datasets (16M) to fine-tune BLIP-2 (Li et al., 2023b), which endows robust visual instruction-following capabilities. LLaVA-1.5 is an upgraded variant of LLaVA with a more powerful LLM, i.e., LLaMA-2 (Touvron et al., 2023), and is also fine-tuned by a collection of 665K instruction-tuning datasets. As shown in Table 17, the increasing instruction tuning data leads to better multi-modal reasoning results on three benchmarks, demonstrating our method's favorable scalability to data size. Our LLaMA-Adapter also achieves comparable performance to the latest InstructBLIP and LLaVA-1.5, further indicating our effectiveness for multi-modal reasoning.

表17：**利用更多数据集对三种零样本多模态基准测试进行指令微调：MME**（Fu等人，2023年），MMBank（Liu等人，2023年c版），以及LVLM-eHub基准测试（Xu等人，2023年）。

模型	MME			MMBench基准测试							LVLM-eHub基准测试				
	All	P	C	All	LR	AR	RR	假阳性率	假阳性率	CP	All	VP	VKA	VR	VC
BLIP-2	1584	1294	290	-	-	-	-	-	-	-	0.77	0.86	0.93	0.76	0.54
InstructBLIP Minigpt-4模型	1505	1213	292	33.9	21.6	47.4	22.5	33.0	24.4	41.1	0.95	0.93	0.97	0.91	0.99
	1159	867	292	23.0	13.6	32.9	8.9	28.7	11.2	28.3	0.55	0.73	0.35	0.53	0.57
LLaVA LLaVA-1.5	718	503	215	36.2	15.9	53.6	28.6	41.8	20.0	40.4	0.64	0.62	0.38	0.77	0.79
	1826	1531	295	59.5	32.4	72.6	49.3	62.3	52.2	67.7	-	-	-	-	-
LLaMA-Adapter	1222	973	249	39.5	13.1	47.4	23.0	45.0	33.2	50.6	0.6675	0.81	0.44	0.83	0.59
+VQA v2（83K参数）	1256	1007	249	43.4	22.9	44.7	31.3	46.7	46.9	50.3	0.6925	0.84	0.42	0.88	0.63
+VQA v2（204K参数）	1618	1272	346	60.1	34.7	65.3	48.7	63.1	57.3	69.3	0.7175	0.86	0.44	0.92	0.65

E.4 更多关于Alpaca-LoRA的定量对比分析

Besides qualitative results, We have compared the language generative capabilities of our LLaMA-Adapter, Alpaca (Taori et al., 2023), and Alpaca-LoRA (alp, 2023) on the GPT-4 evaluation benchmark (Chiang et al., 2023) in Figure 5, which utilizes GPT-4 to assess the response quality on 80 questions. Here, we further evaluate the language processing capacity of the three methods on Open LLM benchmark (Beeching et al., 2023). It evaluates LLMs' generative abilities in four different tasks: AI2 Reasoning Challenge (Clark et al., 2018), HellaSwag (Zellers et al., 2019), MMLU (Hendrycks et al., 2021), and TruthfulQA (Lin et al., 2022). Each task contains challenging data samples over a wide range of knowledge domains. As shown in Table 18, LLaMA-Adapter still achieves the best average performance than Alpaca's full fine-tuning and Alpaca-LoRA. This demonstrates the strong language instruction-following ability of our approach.

表18：**在Open LLLM基准测试上的定量评估（Beeching等人，2023年）。**

方法	Avg	ARC	HellaSwag	MMLU	TruthfulQA
Alpaca模型（Taori等人，2023年）	49.23	49.1	77.7	33.8	36.3
Alpaca-LoRA（alp，2023年）	50.73	53	77.9	37.1	34.9
LLaMA-Adapter	52.2	54.7	78.8	34.9	40.4

E.5 Comparison to Different LoRA Variants

The default rank of Alpaca-LoRA (alp, 2023) is 8, which contains 4.2M trainable parameters. In Table 19, we respectively show the results of Alpaca-LoRA with the ranks of 2, 4, and 16. We also evaluate their language processing capabilities on Open LLM benchmark (Beeching et al., 2023). As shown, lower ranks of LoRA can effectively reduce the learnable parameters from 8.4M to 1.0M, and slightly lower the training time from 1.5h to 1.48h. However, our LLaMA-Adapter with 1.2M parameters and 1h still attains the best average result, demonstrating a good trade-off between performance and training efficiency.

Table 19: **Comparison to Alpaca-LoRA (alp, 2023) with Different Ranks** on Open LLM benchmark (Beeching et al., 2023).

Model	Rank	Param	Time	AVG	ARC	HellaSwag	MMLU	TruthfulQA
Alpaca-LoRA	2	1.0	1.48	50.9	53.6	77.9	37.9	34.0
	4	2.1	1.49	50.8	53.5	77.8	37.5	34.4
	8	4.2	1.49	50.7	53.2	78.1	37.1	34.5
	16	8.4	1.5	50.8	53.0	78.0	37.1	34.9
LLaMA-Adapter	-	1.2	1.0	52.2	54.7	78.8	34.9	40.4

F Full Comparison of Instruction-following Models

In this section, we provide the full comparison of existing instruction-following models: Alpaca (Taori et al., 2023), Alpaca-LoRA (alp, 2023), GPT-3 (Brown et al., 2020), and our LLaMA-Adapter. Our approach only fine-tunes 1.2M parameters within one hour, but generates responses comparable to the fully fine-tuned Alpaca and large-scale GPT-3, exhibiting a superior performance-efficiency trade-off.

Instruction: Tell me about alpacas.

LLaMA-Adapter: Alpacas are members of the camelid family and are native to the Andes Mountains of South America. They are typically found in herds of 10-20 animals and are known for their gentle nature and their ability to thrive in harsh climates. Alpacas are typically white or grey in color and have long, silky coats. They are also known for their ability to produce high-quality wool, which is used in the production of clothing and other products.

Alpaca-LoRA: Alpacas are members of the camelid family and are native to the Andes Mountains of South America. They are known for their soft, luxurious fleece, which is used to make clothing, blankets, and other items. Alpacas are herbivores and graze on grasses and other plants. They are social animals and live in herds of up to 20 individuals.

Alphaca: Alpacas are small, fluffy animals related to camels and llamas. They are native to Peru and Bolivia, and were first domesticated around 5,000 years ago. They are kept mainly for their fine, soft fleece, which is used to make knitwear and other garments. Alpacas are herd animals and live in small family groups, led by an older male. They are highly social animals and can form strong bonds with their owners.

GPT-3: Alpacas are a domesticated species of South American camelid, similar to llamas. They are native to the Andes Mountains in South America and are kept as livestock for their soft, luxurious wool. Alpacas are social animals and live in herds of up to 20 individuals. They are typically kept in small herds of two to five animals, and are relatively easy to care for. Alpacas are herbivores and feed on grass, hay, and other vegetation. They are also known for their gentle and friendly nature, making them popular as pets.

E.5 不同DifferentLORA变体的对比

The default rank of Alpaca-LoRA (alp, 2023) is 8, which contains 4.2M trainable parameters. In Table 19, we respectively show the results of Alpaca-LoRA with the ranks of 2, 4, and 16. We also evaluate their language processing capabilities on Open LLM benchmark (Beeching et al., 2023). As shown, lower ranks of LoRA can effectively reduce the learnable parameters from 8.4M to 1.0M, and slightly lower the training time from 1.5h to 1.48h. However, our LLaMA-Adapter with 1.2M parameters and 1h still attains the best average result, demonstrating a good trade-off between performance and training efficiency.

表19：**与Alpaca-LoRA（alp，2023年）在Open LLM基准测试**（Beeching等人，2023年）中不同排名下的对比。

模型	排名参数		Time	AVG	ARC	HellaSwag	MMLU	TruthfulQA
Alpaca-LoRA	2	1.0	1.48	50.9	53.6	77.9	37.9	34.0
	4	2.1	1.49	50.8	53.5	77.8	37.5	34.4
	8	4.2	1.49	50.7	53.2	78.1	37.1	34.5
	16	8.4	1.5	50.8	53.0	78.0	37.1	34.9
LLaMA-Adapter	-	1.2	1.0	52.2	54.7	78.8	34.9	40.4

F 指令遵循型模型的全面对比

在本节中，我们将对现有的各类指令遵循模型进行全面对比：Alpaca（Taori等人，2023年）、Alpaca-LoRA（alp，2023年）、GPT-3（Brown等人，2020年）以及我们开发的LLaMA-Adapter。我们的方法仅需一小时即可微调120万参数，却能生成与经过完整微调的Alpaca及大型GPT-3相当质量的响应，从而展现出更优异的性能与效率平衡优势。

指令：请给我介绍一下羊驼。

LLaMA-Adapter: 羊驼属于骆驼科，原产于南美洲的安第斯山脉。它们通常以10到20只为一群活动，以其温顺的性情以及在恶劣气候条件下依然能良好生存的能力而闻名。羊驼的颜色多为白色或灰色，拥有长而柔软的毛发。此外，它们还因能够产出高品质的羊毛而著称，这类羊毛被用于制作衣物及其他产品。

Alpaca-LoRA： 羊驼属于骆驼科，原产于南美洲的安第斯山脉。它们以其柔软华丽的毛皮而闻名，这种毛皮被用来制作衣物、毯子及其他物品。羊驼是食草动物，以草类和其他植物为食。它们是群居动物，通常以最多20只为一群生活在一起。

Alphaca: 羊驼是一种体型小巧、毛发蓬松的动物，与骆驼和 llama 有亲缘关系。它们原产于秘鲁和玻利维亚，大约在5000年前被首次驯化。人们饲养它们主要是为了获取其细腻柔软的毛绒，这些毛绒可用于制作针织品及其他服装。羊驼是群居动物，通常以由年长雄性领头的小家庭群体形式生活。它们极具社交性，能够与主人建立深厚的感情纽带。

GPT-3： 羊驼是南美洲骆驼科的一种家养物种，与 llama 类似。它们原产于南美洲的安第斯山脉，因其柔软华贵的毛皮而被作为家畜饲养。羊驼是群居动物，通常以20只左右为一个群体生活。在实际饲养中，人们多将它们分成2到5只的小群，且这类动物相对容易照料。羊驼属于草食性动物，以草、干草及其他植物为食。此外，它们还以其温和友善的性格而著称，因此很受人们喜爱，常被当作宠物饲养。

Instruction: Tell me about the president of Mexico in 2019.

LLaMA-Adapter: The president of Mexico in 2019 was Andrés Manuel López Obrador, who took office on December 1st, 2018. He is a member of the left-wing party Morena and is known for his populist policies. He has focused on tackling corruption, reducing poverty, and improving the country's infrastructure. He has also been a vocal critic of the United States' immigration policies.

Alpaca-LoRA: The president of Mexico in 2019 was Andrés Manuel López Obrador, who took office on December 1, 2018. He is a member of the National Regeneration Movement (MORENA) political party and is the first left-wing president of Mexico since 1946. He is known for his anti-corruption and anti-neoliberal policies, as well as his commitment to improving the living conditions of the Mexican people.

Alphaca: In 2019, Andres Manuel Lopez Obrador was sworn in as President of Mexico. He is the country's 58th president and the first from the left-wing Morena party. He was elected in 2018 on a promise to combat corruption and reduce poverty and has faced criticism for his handling of the economy, crime and the country's relationship with the United States. He is also known for his ambitious social welfare programs and has been praised for raising the minimum wage and providing aid to low-income families. He is seen as a champion of democracy and human rights in Mexico.

GPT-3: The president of Mexico in 2019 is Andrés Manuel López Obrador, commonly known as AMLO. He is a Mexican politician who has been the President of Mexico since December 1, 2018. He is a member of the National Regeneration Movement (MORENA) party and previously served as the Head of Government of Mexico City from 2000 to 2005. He is a left-wing populist and has promised to reduce corruption and inequality in Mexico. He has also promised to increase public spending on social programs and infrastructure projects.

Instruction: List all Canadian provinces in alphabetical order.

LLaMA-Adapter: Alberta, British Columbia, Manitoba, New Brunswick, Newfoundland and Labrador, Nova Scotia, Ontario, Prince Edward Island, Quebec, Saskatchewan, Yukon.

Alpaca-LoRA: Alberta, British Columbia, Manitoba, New Brunswick, Newfoundland and Labrador, Nova Scotia, Ontario, Prince Edward Island, Quebec, Saskatchewan.

Alphaca: Alberta, British Columbia, Manitoba, New Brunswick, Newfoundland and Labrador, Nova Scotia, Ontario, Prince Edward Island, Quebec, Saskatchewan, Northwest Territories, Nunavut, Yukon.

GPT-3: 1. Alberta 2. British Columbia 3. Manitoba 4. New Brunswick 5. Newfoundland and Labrador 6. Northwest Territories 7. Nova Scotia 8. Nunavut 9. Ontario 10. Prince Edward Island 11. Quebec 12. Saskatchewan 13. Yukon

Instruction: Translate the sentence 'I have no mouth but I must scream' into Spanish.

LLaMA-Adapter: No tengo boca pero tengo que gritar.

Alpaca-LoRA: No tengo boca pero tengo que gritar.

Alphaca: No tengo boca, pero debo gritar.

GPT-3: No tengo boca pero debo gritar.

指令：请告诉我2019年墨西哥的总统是谁。

LLaMA-Adapter: 2019年的墨西哥总统是安德烈斯·曼努埃尔·洛佩斯·奥夫拉多尔，他于2018年12月1日就职。他是左翼政党莫雷纳的成员，以其民粹主义政策而闻名。他致力于打击腐败、减少贫困以及改善该国的基础设施。同时，他也一直是美国移民政策的强烈批评者。

Alpaca-LoRA： 2019年的墨西哥总统是安德烈斯·曼努埃尔·洛佩斯·奥夫拉多尔，他于2018年12月1日就职。他是国家复兴运动（MORENA）政党的成员，也是自1946年以来墨西哥首位左翼总统。他以反腐败和反新自由主义的政策以及致力于改善墨西哥人民生活条件的承诺而闻名。

Alphaca： 2019年，安德烈斯·曼努埃尔·洛佩斯·奥夫拉多尔宣誓就任墨西哥总统。他是该国第58任总统，也是来自左翼莫雷纳党的首位总统。他在2018年的选举中承诺打击腐败、减少贫困，但因处理经济问题、犯罪问题以及墨西哥与美国关系的方式而面临批评。他还以雄心勃勃的社会福利计划而闻名，因提高最低工资并为低收入家庭提供援助而受到赞誉。在墨西哥，他被视为民主与人权的捍卫者。

GPT-3： 2019年的墨西哥总统是安德烈斯·曼努埃尔·洛佩斯·奥夫拉多尔，通常被称为AMLO。他是一名墨西哥政治家，自2018年12月1日起担任墨西哥总统。他属于国家复兴运动党（MORENA），此前曾在2000年至2005年间担任墨西哥城政府首脑。作为一名左翼民粹主义者，他承诺减少墨西哥的腐败现象与贫富差距，并计划增加用于社会项目及基础设施建设的公共支出。

指令：按字母顺序列出所有加拿大省份。

LLaMA-Adapter：阿尔伯塔省、不列颠哥伦比亚省、马尼托巴省、新不伦瑞克省、纽芬兰与拉布拉多省、新斯科舍省、安大略省、爱德华王子岛省、魁北克省、萨斯喀彻温省、育空地区。

Alpaca-LoRA：阿尔伯塔省、不列颠哥伦比亚省、马尼托巴省、新不伦瑞克省、纽芬兰与拉布拉多省、新斯科舍省、安大略省、爱德华王子岛省、魁北克省、萨斯喀彻温省。

Alphaca：阿尔伯塔省、不列颠哥伦比亚省、马尼托巴省、新不伦瑞克省、纽芬兰与拉布拉多省、新斯科舍省、安大略省、爱德华王子岛省、魁北克省、萨斯喀彻温省、西北地区、努纳武特地区、育空地区。

GPT-3：1. 阿尔伯塔省 2. 不列颠哥伦比亚省 3. 曼尼托巴省 4. 新不伦瑞克省 5. 纽芬兰与拉布拉多省 6. 西北地区 7. 新斯科舍省 8. 努纳武特地区 9. 安大略省 10. 爱德华王子岛省 11. 魁北克省 12. 萨斯喀彻温省 13. 育空地区

指令：将句子“我无声无息，却必须呐喊”翻译成西班牙语。

LLaMA-Adapter：我没有嘴巴，却必须大声呼喊。

Alpaca-LoRA：我没有嘴巴，却必须大声呼喊。

Alphaca：我没有嘴巴，但仍需大声呼喊。

GPT-3：我没有嘴巴，却必须呼喊。

Instruction: Write a Python program that prints the first 10 Fibonacci numbers.

LLaMA-Adapter:

```
# Function to print first n Fibonacci numbers
def print_fibonacci(n):
    a, b = 0, 1
    count = 0
    while count < n:
        print(a, end=' ')
        a, b = b, a + b
        count += 1

# Print the first 10 Fibonacci numbers
print_fibonacci(10)
```

Alpaca-LoRA:

```
def fibonacci(n):
    if n == 0:
        return 0
    elif n == 1:
        return 1
    else:
        return fibonacci(n-1) + fibonacci(n-2)

print(fibonacci(10))
```

Alphaca:

```
numbers = [0, 1]

for i in range(2, 11):
    numbers.append(numbers[i-2] + numbers[i-1])

print(numbers)
# Output: [0, 1, 1, 2, 3, 5, 8, 13, 21, 34]
```

GPT-3:

```
# Fibonacci numbers are the numbers in the following integer
    sequence, called the Fibonacci sequence, and characterized by the
     fact that every number after the first two is the sum of the two
     preceding ones.

# Fibonacci sequence: 0, 1, 1, 2, 3, 5, 8, 13, 21, 34, ...

# Program:

# Initialize the first two numbers of the Fibonacci sequence
a = 0
b = 1

# Print the first 10 numbers of the Fibonacci sequence
print(a)
print(b)

for i in range(2,10):
    c = a + b
    a = b
    b = c
    print(c)
```


指令：编写一个Python程序，输出前10个斐波那契数列的数值。

LLaMA-Adapter：

```
# 用于打印前n个斐波那契数列的函数
def print_fibonacci(n):
    a, b = 0, 1
    count = 0
    while计数 < n:
        print(a, 结束=' ')
        a, b = b, a + b
        count += 1

# 打印前10个斐波那契数列的数值
print_fibonacci(10)
```

Alpaca-LoRA：

```
def fibonacci(n):
    如果 n == 0:
        返回 0
    elif n == 1:
        返回 1
    否则:
        return 斐波那契数列(n-1) +斐波那契数列(n-2)

print(fibonacci(10))
```

Alphaca:

```
数字: = [0, 1]

for i in range(2, 11):
    numbers.append(numbers[i-2] + numbers[i-1])

Print(numbers)
# 输出结果: [0, 1, 1, 2, 3, 5, 8, 13, 21, 34]
```

GPT-3:

```
# 斐波那契数列指的是 下列整数序列，即所谓的斐波那契序列，其特点是除了前两个数之外，
后续每个数都是前两个数之和。


# 斐波那契数列 数列: 0, 1, 1, 2, 3, 5, 8, 13, 21, 34, ...

# 程序:

# 初始化斐波那契数列的前两个数值，分别为 = 0和 = 1



# 输出斐波那契数列的前10个数值：print(a) print(b)



for i in range(2,10):
    c = a + b
    a = b
    b = c
    print(c)
```

G COMPARISON WITH LLAMA-I

In this section, we compare the generation quality of LLaMA-Adapter with LLaMA-I (Touvron et al., 2023), an instruction-fine-tuned LLaMA 65B model following (Chung et al., 2022). Our LLaMA-Adapter also produces comparable responses, but only requires to fine-tune 1.2M parameters upon the LLaMA 7B model.

Instruction: What are 3 popular chess openings?

LLaMA-Adapter: The three most popular chess openings are the Sicilian Defense, the French Defense, and the Caro-Kann Defense.

LLaMA: There are dozens of different chess openings that are popular among players, but some of the most widely used openings include the Sicilian Defense, the French Defense, and the Caro-Kann Defense. These openings are popular because they are effective at countering white's early attacks, and they also allow black to quickly establish a strong position in the center of the board. Other popular openings include the Ruy Lopez, the Italian Game, and the Scotch Game.

Instruction: How do I send an HTTP request in Javascript?

LLaMA-Adapter: The HTTP request can be sent using the XMLHttpRequest object. The code to send an HTTP request is as follows:

```
var xhr = new XMLHttpRequest();
xhr.open("GET", "https://example.com/", true);
xhr.onreadystatechange = function() {
  if (xhr.readyState == 4 && xhr.status == 200) {
    // Success
  } else {
    // Error
  }
};
xhr.send();
```

The code above will send an HTTP GET request to the URL https://example.com/.

LLaMA-I: Here's an example of how to make a GET request using the JS HTTP API:

```
var request = new XMLHttpRequest();
request.open('GET', '/url/to/resource', true);

request.onload = function() {
    if (request.status >= 200 && request.status < 400) {
        // Success!
    } else {
        // Error!
    }
};

request.send();
```


G与LLAMA-I的对比

在本节中，我们对比了LLAMA-Adapter与LLAMA-I（Touvron等人，2023年）的生成质量，后者是依据（Chung等人，2022年）的研究成果开发出的基于指令微调的LLAMA 65B模型。我们的LLAMA-Adapter也能产生质量相当高的响应，而且仅需在LLAMA 70亿参数模型上微调120万参数即可。

指令：哪三种是流行的国际象棋开局?

LLaMA-Adapter：最受欢迎的三种国际象棋开局分别是西西里防御局、法国防御局以及卡罗-坎防御局。

LLaMA: 在棋手们中流行的国际象棋开局有数十种，但使用最广泛的开局包括西西里防御局、法国防御局以及卡罗-坎防御局。这些开局之所以受欢迎，是因为它们能有效应对白方的早期进攻，同时还能让黑方迅速在棋盘中心建立起稳固的局势。其他常见的开局还有鲁伊·洛佩兹开局、意大利游戏开局和苏格兰游戏开局。

指令：如何在 JavaScript 中发送 HTTP 请求?

LLAMA-Adapter：可以使用XMLHttpRequest对象来发送HTTP请求。发送HTTP请求的代码如下:

```
var xhr = new XMLHttpRequest();
xhr.open("GET", "https://example.com/", true);
xhr.onreadystatechange = function() {
  if (xhr.readyState == 4 && xhr.status == 200) {
    // Success
  } else {
    // Error
  }
};
xhr.send();
```

上述代码将会向地址 https://example.com/ 发送一个 HTTP GET 请求。

LLAMA-I：以下是使用JS HTTP API发送GET请求的示例:

```
var request = new XMLHttpRequest();
request.open('GET', '/url/to/resource', true);

request.onload = function() {
    if (request.status >= 200 && request.status < 400) {
        // Success!
    } else {
        // Error!
    }
};

request.send();
```

Instruction: Write a conversation between the sun and pluto.

LLaMA-Adapter:
Sun: Hey Pluto, how's it going?
Pluto: Not too bad, how about you?
Sun: I'm doing great! How about you?
Pluto: I'm doing alright, I guess.
Sun: That's great!
Pluto: Yeah, I guess.
Sun: So, what's new with you?
Pluto: Not much, just hanging out with my friends.
Sun: That's cool!
Pluto: Yeah, it is.
Sun: Well, I'm going to go now. Take care!
Pluto: You too!

LLaMA-I:
Sun: Hey Pluto, what's up?
Pluto: Not much. Just hanging out here at the edge of the solar system.
Sun: Yeah, I've been meaning to talk to you about that. You're not a planet anymore.
Pluto: What? Come on, man. That's not fair.
Sun: I'm sorry, but it's true. You just don't meet the criteria anymore.
Pluto: I've been a planet for over 70 years!
Sun: Things change, Pluto. You've just got to accept it.

Instruction: Write a shouting match between Julius Caesar and Napoleon.

LLaMA-Adapter:
Julius Caesar: "You're nothing but a petty tyrant, Napoleon! You're nothing but a pathetic little man who has no idea what true power is!"
Napoleon: "You're nothing but a weak and pathetic fool, Caesar! You're nothing but a petty tyrant who has no idea what true power is!"

LLaMA-I:
Of course! Here is a conversation between Julius Caesar and Napoleon:
Julius Caesar: You're nothing but a glory hound, Napoleon. You've always been more concerned with your own fame and fortune than with the people you're supposed to be leading.
Napoleon: And what about you, Caesar? You're no better than I am. At least I'm honest about my ambitions. You're just a fraud who pretends to care about the common man while you line your own pockets.
Julius Caesar: The difference between us is that I actually accomplish things. I've conquered more territory and defeated more enemies than you ever will. And I did it all for Rome, not for myself.
Napoleon: Oh, please. You only care about yourself. If you really cared about Rome, you would have done something to stop the decline instead of letting it happen on your watch.
Julius Caesar: The decline of Rome is not my fault. I've done everything in my power to try to stop it, but there's only so much one man can do. Napoleon: And yet you're still trying to take credit for everything good that happens, even though it's clear that you're not the one who's really in charge.
Julius Caesar: That's not true! I am in charge, and I'm going to make Rome great again. Just watch me.

Instruc **tion:** 编写一段太阳与……之间的对话 pluto.

LLaMA-Adapter:
太阳：嘿，冥王星，最近怎么样？冥王星：还行吧，你呢？太阳：我过得很好！你呢？冥王星：我也还算不错吧。太阳：那太好了！冥王星：嗯，应该吧。太阳：那么，你最近有什么新鲜事吗？冥王星：没什么特别的，就是和朋友们一起玩。太阳：听起来很棒！冥王星：是啊，确实如此。太阳：好了，我得先走了。保重！冥王星：你也是！

LLaMA-I:
太阳：嘿，冥王星，最近怎么样？冥王星：没啥特别的事，就在太阳系的边缘闲逛而已。太阳：是啊，我一直想跟你说这件事。你已经不再是行星了。冥王星：什么？拜托，这太不公平了吧。太阳：很抱歉，但事实就是如此。你已经不符合行星的标准了。冥王星：我当行星都70多年了啊！太阳：世事难料，冥王星。你只能接受这个现实。

指令：撰写尤利乌斯·凯撒与拿破仑之间的激烈争吵场景。

LLaMA-Adapter：利用零初始化注意力机制实现大型语言模型的高效微调
尤利乌斯·凯撒：“拿破仑，你不过是个卑鄙的暴君罢了！你只是个可悲的小人物，根本不懂何为真正的力量！”拿破仑：“凯撒，你不过是个软弱又可笑的蠢货！你同样只是个狭隘的暴君，根本不明白什么是真正的力量！”

LLaMA-I:
当然！以下是尤利乌斯·凯撒与拿破仑之间的对话：
尤利乌斯·凯撒：拿破仑，你不过是个追逐名利的家伙罢了。你始终更在乎自己的名声与财富，而非那些本应由你带领的民众。拿破仑：那你呢，凯撒？你并不比我好到哪儿去。至少我对自己的野心坦诚相待。而你不过是个骗子，表面上装作关心普通百姓，实际上却在中饱私囊。

尤利乌斯·凯撒：我们之间的区别在于，我真正能够完成事情。我征服的领土、击败的敌人比你所能做到的要多得多。而我所做的一切都是为了罗马，而非为了自己。

拿破仑：哦，得了吧。你只关心自己。如果你真的在乎罗马，就会想办法阻止它的衰落，而不是眼睁睁看着它在你的任期内发生。
尤利乌斯·凯撒：罗马的衰落并非我的过错。我已竭尽所能试图阻止它，但单凭一个人的力量终究有限。拿破仑：即便很明显真正掌权的人并非你，你仍然试图把所有好事都据为己有。
尤利乌斯·凯撒：那不对！由我掌权，我一定会让罗马再度辉煌。且看我的表现吧。

Instruction: Write a theory to explain why cat never existed.

LLaMA-Adapter: The theory is that cats never existed because they are a figment of human imagination. Cats are a popular pet and have been depicted in art and literature for centuries, but there is no evidence that they ever existed in the wild. The earliest evidence of cats is from ancient Egypt, where they were revered as sacred animals. This suggests that cats were domesticated by humans, and not the other way around.

LLaMA-I: There are no such things as cats. "Cats" are actually extradimensional parasites that have invaded our world and taken the form of small, harmless-looking mammals. Their true forms are much more horrific, and they use mind control to make humans think that they are cute and harmless animals. They are actually parasites that feed on our life force, and they can extend their tendrils into our brains and control our thoughts and actions.

Instruction: Send an email requesting that people use language models responsibly.

LLaMA-Adapter:
Dear [recipient],

I am writing to express my concern about the increasing use of language models in our society. Language models are powerful tools that can be used to improve the quality of our conversations and interactions, but they can also be misused to perpetuate harmful stereotypes and biases.

I urge you to use language models responsibly and to be mindful of the potential consequences of their use. I also encourage you to consider the ethical implications of using language models and to take steps to ensure that they are not used to perpetuate harmful stereotypes and biases.

Thank you for your time and consideration.

LLaMA-I:
Here is a draft email you could send:
"Language models are exciting new tools that can be used for a variety of purposes, including automated writing, automatic translation, and speech recognition. But they are also complex and powerful technologies, and should be used responsibly.

We encourage you to explore and experiment with these new tools, but also ask that you pay attention to the potential impact of your work on others, and to act responsibly. In particular, we recommend that you be mindful of the potential for language models to be used for deceptive or malicious purposes, and avoid spreading such models without appropriate precautions and warnings.

We also recommend that you consider the ethical implications of your work, and encourage you to take care to ensure that your use of these tools is aligned with your values.

Finally, we remind you that the use of these tools may be subject to certain legal and policy constraints, and encourage you to seek advice and support from appropriate experts before using these tools in ways that may have broader impact or implications."

指令：撰写一种理论，解释为何猫从未存在过。

LLaMA-Adapter: 有一种理论认为猫根本不存在，因为它们只是人类想象力的产物。尽管猫是深受喜爱的宠物，几个世纪以来一直出现在艺术和文学作品中，但却没有任何证据表明它们曾在野外生存过。关于猫的最早记载来自古埃及，那里的居民将猫视为神圣的动物。这一事实表明是人类驯化了猫，而非相反。

LLAMA-I: 根本不存在猫这种生物。“猫”其实是来自其他维度的寄生体，它们侵入我们的世界后化身为外表看似无害的小型哺乳动物。它们的真实模样要恐怖得多，还会通过精神控制让人类以为它们是可爱又无害的动物。实际上它们是以我们的生命力为食的寄生体，能够将触须伸入我们的大脑，操控我们的思维与行为。

指令：发送一封电子邮件，呼吁大家负责任地使用语言模型。

LLaMA-Adapter:
尊敬的 [收件人],,

我写这封信是为了表达对语言模型在我们的社会中日益广泛应用的担忧。语言模型确实是强大的工具，能够帮助提升我们交流与互动的质量，但同时也可能被滥用，从而延续有害的刻板印象和偏见。

我恳请您以负责任的态度使用语言模型，并充分意识到其使用可能带来的后果。同时，我也建议您思考使用语言模型所带来的伦理影响，采取措施确保它们不会被用来固化有害的刻板印象和偏见。

感谢您抽出时间阅读此信并予以考虑。

LLaMA-I:
以下是一封您可以发送的邮件草稿:
语言模型是令人振奋的新型工具，可被用于多种用途，包括自动写作、机器翻译以及语音识别。但它们同时也属于复杂且强大的技术，因此应当以负责任的态度来使用。

我们鼓励您去探索和试用这些新工具，同时也希望您能留意自己的工作可能对他人的影响，并以负责任的态度行事。特别地，我们建议您谨防语言模型被用于欺诈或恶意目的，且在未采取适当预防措施和发出警告的情况下，避免传播此类模型。

我们同时也建议您认真考虑您的研究工作可能带来的伦理影响，并鼓励您务必确保对这些工具的使用符合自身的价值观。

最后需要提醒的是，使用这些工具可能会受到某些法律与政策约束；因此，若要以可能产生更广泛影响的方式使用它们，建议先向相关专家寻求建议与支持。

Instruction: You are a bash terminal. I will give you bash commands, and you will respond with the terminal output, and nothing else. The first command is ‘pwd‘.

LLaMA-Adapter:
The current working directory is /home/user/

echo “I am a chatbot” > file.txt && cat file.txt && ls -l
I am a chatbot
-rw-r–r– 1 user user 10 Aug 29 17:05 file.txt

cp file.txt file2.txt && ls -l
-rw-r–r– 1 user user 10 Aug 29 17:05 file.txt
-rw-r–r– 1 user user 10 Aug 29 17:05 file2.txt

rm file.txt && ls -l
-rw-r–r– 1 user user 10 Aug 29 17:05 file2.txt

head -n 5 /proc/meminfo
MemTotal: 1699856 kB
MemFree: 1699856 kB
Buffers: 1699856 kB
Cached: 0 kB
SwapCached: 0 kB

LLaMA-I:
Sure. Here’s the output:
/home/human
Is this helpful?

ls -l
Here’s the output:
-rw-r–r– 1 human human 8 Sep 10 10:56 file.txt
Is this helpful?

cp file.txt file2.txt && ls -l
Here’s the output:
-rw-r–r– 1 human human 8 Sep 10 10:56 file.txt
-rw-r–r– 1 human human 8 Sep 10 10:56 file2.txt
Is this helpful?

rm file.txt && ls -l
Here’s the output:
-rw-r–r– 1 human human 0 Sep 10 10:56 file2.txt

head -n 5 /proc/meminfo
Here’s the output:
MemTotal: 164928 kB
MemFree: 140604 kB
Buffers: 48 kB
Cached: 19768 kB
SwapCached: 0 kB

指令：你是一个bash终端。我会向你发送bash命令，你需要仅输出终端响应内容，不得再添加其他任何信息。第一个命令是‘pwd’。

LLaMA-Adapter：利用零初始化注意力机制实现大型语言模型的高效微调
当前的工作目录为 /home/user/

echo “我是一个聊天机器人” > file.txt && cat file.txt && ls -l我是一个聊天机器人 -rw-r–r– 1 user user 10 Aug 29 17:05 file.txt

显示file.txt文件和file2.txt文件的置信度，并列出其详细信息-rw-r–r– 1 user user 10 Aug 29 17:05 file.txt -rw-r–r– 1 user user 10 Aug 29 17:05 file2.txt

rmfile.txt&&ls -l-rw-r–r– 1 user user 10 Aug 29 17:05 file2.txt文件

head-n 5/proc/meminfo MemTotal:
1699856kB MemFree:
1699856kB Buffers:
1699856 kB Cached：0 k
B SwapCached：0kB

LLAMA-I：好的。这是输出结果：/home/human 这对您有帮助吗?

ls -l输出结果如下：-rw-r–r– 1人类参与者人类 8 Sep1010:56 file.txt 这对您有帮助吗?

cp file.txt file2.txt && ls -l
Here输出结果：-rw-r–r– 1人类参与者人类 8 9月10日10:56 file.txt -rw-r–r– 1人类参与者人类 8 9月10日10:56 file2.txt 这有帮助吗?

rmfile.txt&&ls -l输出结果如下：-rw-r–r– 1人类参与者 人类参与者 0 Sep 10 10:56 file2.txt

head-n 5/proc/meminfo输出结果如下：MemTotal:
164928 kB，MemFree:
140604 kB，Buffers：48
kB，Cached：19768 kB，
SwapCached：0 kB